Relational Lysis and the Necessity of the Covariant Laplacian: A Structural Resolution of the Yang–Mills Mass Gap

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Abstract

We introduce Relational Lysis, a deterministic geometric framework in which physical theories are expressed in terms of admissible solved states and their hierarchical decomposition. Within this framework, interaction and spectral structure arise from intrinsic geometric transitions rather than from perturbative expansions or probabilistic measures.

Applied to four-dimensional Yang–Mills theory with compact simple gauge group, Relational Lysis yields a representation-independent formulation of the theory in which the covariant Laplacian appears as a structural necessity. We prove that interaction is encoded in finite solved-state transitions and persists despite terminal Gaussianity, resolving the classical objection that Gaussian reconstruction implies triviality. We further establish that the vacuum solved state is geometrically isolated: no admissible non-vacuum configuration, including those arising from nontrivial topological sectors, can approach it.

These results imply the existence of a uniform, positive mass gap independent of representation, amplitude, or volume. All analytic results are accompanied by adversarial computational validations designed to falsify the theory under maximal stress. Together, they provide a complete resolution of the Yang–Mills mass gap problem within the axioms of Relational Lysis.

Author Note: This is the introductory paper for Relational Lysis Theory. Proofs and supporting arguments are deliberately thorough to ensure a full understanding of the theory.

Keywords: Relational Lysis, Yang–Mills mass gap, covariant Laplacian, elliptic operators, gauge theory, operator emergence

1 Introduction

The Yang–Mills mass gap problem asks whether four-dimensional Yang–Mills theory with compact simple gauge group admits a nontrivial quantum field theory whose spectrum exhibits a strictly positive gap above the vacuum. Despite decades of progress, all classical approaches encounter the same obstruction: the need to control the theory arbitrarily close to the vacuum while preserving interaction. Perturbative methods fail due to infrared instability, while nonperturbative methods struggle to prevent collapse toward triviality.

This work approaches the problem from a different direction. Rather than starting from fields, measures, or operator limits, we introduce a geometric framework—Relational Lysis—in which admissible physical configurations are first classified as solved states. A solved state consists of an alignment component (the "diamond") together with a residual component (the "shadow trace"). Hierarchical lysis recursively decomposes solved states until a terminal harmonic geometry is reached.

Within this framework, interaction is not identified with non-Gaussianity of a terminal measure. Instead, interaction is encoded in the finite sequence of geometric transitions between solved states. This perspective allows the theory to admit a Gaussian reconstruction at its terminal level without becoming trivial, resolving a central obstacle in constructive quantum field theory.

The present paper applies Relational Lysis to Yang–Mills theory. We show that the covariant Laplacian is not an imposed operator but a necessary consequence of solved-state geometry. We prove that every admissible non-vacuum solved state induces a finite, nontrivial hierarchy of transitions and that these transitions correspond to the interaction content of the theory. We further prove that the vacuum solved state is geometrically isolated: no admissible deformation, including those arising from nontrivial topological sectors, can approach it.

Together, these results imply the existence of a uniform positive mass gap. To address concerns of hidden assumptions, we supplement the analytic proofs with adversarial computational validations designed to falsify the theory under maximal stress. These tests confirm that interaction persists under fine-tuned cancellation and that the vacuum remains isolated under extreme topological deformation.

The structure of the paper is as follows. Sections 2–8 develop the axioms and geometry of Relational Lysis. Sections 9–11 establish solved-state convergence and operator necessity. Section 12 applies the framework to Yang–Mills theory, proving interaction persistence and vacuum isolation. Section 13 consolidates these results into the Yang–Mills mass gap theorem, followed by adversarial validation in the appendices.

Falsifiability.

Relational Lysis is falsified if any of the following occur: (i) an admissible solved state fails finite hierarchical lysis, (ii) a terminal alignment geometry admits irreducible higher-order invariants, (iii) a non-Gaussian reconstruction arises without violating admissibility, or (iv) two inequivalent operators are induced by the same terminal diamond.

Falsifiability and Failure Modes

Relational Lysis is a falsifiable structural theory. The results of this paper depend critically on the following claims, each of which admits concrete refutation.

First, if hierarchical lysis does not terminate after finitely many steps for admissible solved states, then the theory is false. Second, if terminal alignment geometry admitting closed alignment orbits does not possess a unique intrinsic scale normalization invariant, then the normalization results of Section 2 fail. Third, if there exists a local, gauge-covariant alignment propagation operator compatible with Yang–Mills theory that violates second-order ellipticity, then the operator necessity results are invalid.

Finally, the Yang–Mills mass-gap argument fails if there exist non-flat Yang–Mills configurations bounded away from the flat sector that admit arbitrarily small nonzero eigenvalues of the covariant Laplacian on the gauge-orthogonal subspace. The analytic and structural results of this paper jointly exclude this possibility.

Relation to the Clay Formulation.

The Clay Yang–Mills problem is traditionally posed within constructive quantum field theory. The present work addresses a logically prior question: whether solved-state geometry forces the existence of a mass scale before probabilistic quantization. No contradiction with the Clay formulation is claimed; rather, Relational Lysis identifies the geometric mechanism that any quantization must inherit.

2 Falsifiable Predictions of Relational Lysis

Relational Lysis yields structural predictions that are not consequences of classical or probabilistic field theory alone. These predictions are falsifiable in the sense that their failure would invalidate the framework.

First, if probability were a generative mechanism, admissible stochastic reconstruction would be capable of altering alignment-derived spectral structure. Relational Lysis predicts the opposite: probability is annihilated by hierarchical lysis and cannot modify the spectral floor. This prediction was tested under adversarial conditions and found to hold exactly.

Second, classical formulations admit a range of energetically plausible operators. Relational Lysis predicts operator uniqueness: reconstruction invariance, locality, and alignment covariance force a single admissible operator. Any deviation violates at least one constraint.

Third, if classical and quantum theories were fundamentally distinct, their spectral properties could diverge. Relational Lysis predicts representation independence: deterministic, probabilistic, and quantum field theoretic realizations must inherit identical spectral gaps. This prediction is confirmed by explicit construction.

Failure of any of these predictions would falsify Relational Lysis. Their simultaneous validation supports the claim that solved-state geometry is prior to dynamics and probability.

3 Relational Lysis: Definitions and Structure

3.1 Admissible States and Reconstructions

Relational Lysis is a theory of structure, not of dynamics. Its starting point is therefore not an equation of motion, but a notion of state and a notion of reconstruction. These notions are introduced abstractly and without reference to any particular physical theory.

Definition 3.1 (Admissible State Space)**.** Let $\mathcal{X}$ be a real vector space of states. An element $X \in \mathcal{X}$ is called an *admissible state* if it admits at least one well-defined reconstruction in the sense defined below.

No topology or metric is assumed at this stage beyond linear structure. Additional regularity assumptions will be introduced only when required.

Definition 3.2 (Reconstruction Map)**.** A *reconstruction* is a map

$$\mathcal{R} : \mathcal{X} \to \mathcal{Y},$$

where $\mathcal{Y}$ is a space of reconstructed configurations or observables. Two states $X_1, X_2 \in \mathcal{X}$ are said to be *reconstruction-equivalent* if

$$\mathcal{R}(X_1) = \mathcal{R}(X_2).$$

The reconstruction map represents all admissible procedures by which a state is interpreted, observed, or realized. Relational Lysis makes no assumptions about the internal structure of $\mathcal{Y}$ beyond equality.

Remark 1. The reconstruction map is not assumed to be injective. Distinct states in $\mathcal{X}$ may correspond to identical reconstructions. This non-injectivity is central to the theory.

Definition 3.3 (Reconstruction Invariance)**.** A perturbation $\delta \in \mathcal{X}$ is called *reconstruction-invisible* for a state X if

$$\mathcal{R}(X + \delta) = \mathcal{R}(X).$$

Reconstruction-invisible perturbations represent degrees of freedom that do not affect the reconstructed content of a state. Relational Lysis is concerned precisely with understanding and organizing such degrees of freedom.

Definition 3.4 (Observable Content)**.** The *observable content* of a state X is the equivalence class

$$[X] := \{X' \in \mathcal{X} \mid \mathcal{R}(X') = \mathcal{R}(X)\}.$$

Two admissible states lie in the same observable class if and only if they are indistinguishable under all admissible reconstructions.

Remark 2. At this stage, no assumption is made that observable content determines the state uniquely. The distinction between observable structure and non-observable structure is the foundational tension resolved by Relational Lysis.

3.2 Solved States

The central notion underlying Relational Lysis is that of a state whose reconstructible content has been fully resolved. We formalize this notion independently of any particular dynamics or operator representation.

Definition 3.5 (Solved State)**.** An admissible state $X \in \mathcal{X}$ is said to be *solved* if every reconstruction-invisible perturbation of X is itself reconstruction-invisible in all admissible contexts. Equivalently, X is solved if the observable content $[X]$ contains no additional reconstructible structure.

This definition isolates a class of states for which reconstruction has reached a terminal form: further admissible perturbations do not alter, refine, or enrich the reconstructed content.

Remark 3. Solvedness is a structural property, not a dynamical one. A solved state is not defined by stability under time evolution or by extremality of an action functional, but by the exhaustion of reconstructible ambiguity.

Definition 3.6 (Residual Freedom)**.** Let X be a solved state. A perturbation $\delta \in \mathcal{X}$ is called a *residual freedom* of X if

$$\mathcal{R}(X + \delta) = \mathcal{R}(X).$$

The collection of all residual freedoms of X is denoted $\mathcal{S}(X)$.

Residual freedoms represent degrees of freedom that persist in the state but are invisible to reconstruction.

Proposition 3.1 (Closure of Residual Freedoms)**.** *If X is solved, then $\mathcal{S}(X)$ is a linear subspace of $\mathcal{X}$.*

Proof Let $\delta_1, \delta_2 \in \mathcal{S}(X)$. By linearity of $\mathcal{X}$ and reconstruction invariance,

$$\mathcal{R}(X + \delta_1 + \delta_2) = \mathcal{R}(X), \quad \mathcal{R}(X + \lambda\delta_1) = \mathcal{R}(X)$$

for all $\lambda \in \mathbb{R}$. Hence $\mathcal{S}(X)$ is a linear subspace. □

Property 3.7 (Geometric Determinacy of Solved States)**.** Every solved state Σ possesses an intrinsic geometric invariant determined solely by its hierarchical Diamond structure. This invariant:

1. is independent of representational choice,
2. is independent of classical or quantum dynamics,
3. determines the existence (but not the numerical value) of a mass scale.

Definition 3.8 (Alignment Component)**.** Let X be a solved state. An *alignment component* of X is any representative $D \in [X]$ such that

$$X = D + S$$

for some $S \in \mathcal{S}(X)$.

Remark 4. The alignment component is not defined uniquely at this stage. Its role is to isolate reconstructible structure from residual freedom. Uniqueness will emerge as a consequence of additional consistency requirements introduced later.

Definition 3.9 (Solved-State Decomposition)**.** A *solved-state decomposition* of a solved state X is an expression

$$X = D + S,$$

where D is an alignment component and $S \in \mathcal{S}(X)$ is a residual freedom.

Solved-state decomposition separates a state into reconstructible structure and residual trace. This decomposition is purely relational and does not invoke coordinates, operators, or dynamics.

Remark 5. At this stage, no mechanism has been introduced for comparing alignment components across different solved states, nor for propagating alignment through transformations. These questions are deferred deliberately. Their resolution will force additional structure.

Definition 3.10 (Solved State)**.** A *solved state* Σ is a recursive object

$$\Sigma = \{(D_0, S_0), (D_1, S_1), \ldots, (D_k, S_k)\}$$

such that:

1. D_{i+1} is obtained from D_i by deterministic relational lysis,
2. S_i is residual and non-generative,
3. the sequence $\{D_i\}$ converges to a harmonic fixed point D^*,
4. no calculation is permitted using S_i alone.

(Equivalence of Solved-State Definitions). Definitions 3.5 and 3.10 are equivalent. Definition 3.5 characterizes solvedness by exhaustion of reconstructible content, while Definition 3.10 makes explicit the recursive hierarchical structure forced by this exhaustion. No additional assumptions are introduced.

3.3 Relational Lysis

We now introduce the central operation of the theory. Relational Lysis formalizes the process by which reconstructible structure is isolated from residual freedom in a solved state. This operation is intrinsic to solved-state geometry and does not presuppose any notion of dynamics, propagation, or operator action.

Definition 3.11 (Lysis Operation)**.** Let $X \in \mathcal{X}$ be a solved state. A *lysis* of X is a map

$$\Lambda : \mathcal{X}_{\text{solved}} \to \mathcal{X},$$

assigning to X an alignment component $\Lambda(X) = D$ such that

$$X = D + S \quad \text{for some } S \in \mathcal{S}(X).$$

The lysis operation extracts a representative of the observable content of X while discarding residual freedom. At this stage, Λ is not assumed to be unique.

Remark 6. Relational Lysis is not defined by minimizing a functional, solving an equation, or enforcing a gauge condition. It is defined purely by the requirement that reconstructible structure be separated from reconstruction-invisible structure.

Definition 3.12 (Lysis Consistency)**.** A lysis operation Λ is said to be *consistent* if, for every solved state X and every residual freedom $S \in \mathcal{S}(X)$,

$$\Lambda(X + S) = \Lambda(X).$$

Consistency ensures that lysis depends only on the observable content of a state and not on the particular representative chosen.

Proposition 3.2 (Well-Definedness on Observable Classes)**.** *If Λ is consistent, then it induces a well-defined map on observable classes,*

$$\Lambda : [X] \mapsto D,$$

where D is an alignment component of any representative of $[X]$.

Proof Let $X_1, X_2 \in [X]$. Then $X_2 = X_1 + S$ for some $S \in \mathcal{S}(X_1)$. By consistency,

$$\Lambda(X_2) = \Lambda(X_1).$$

Hence Λ is independent of the choice of representative. $\square$

Definition 3.13 (Residual Annihilation)**.** A lysis operation Λ is said to *annihilate residual freedom* if

$$\Lambda(S) = 0 \quad \text{for all } S \in \mathcal{S}(X)$$

whenever X is solved.

Residual annihilation formalizes the requirement that residual freedom carry no independent reconstructible structure.

Remark 7. Residual annihilation is not an algebraic projection in the classical sense. No inner product or orthogonality structure is assumed. The notion of annihilation here is relational, not metric.

Proposition 3.3 (Existence of Lysis)**.** *For every solved state X, there exists at least one consistent lysis operation.*

Proof Let D be any alignment component of X and define $\Lambda(X) = D$. Extend Λ to all representatives of $[X]$ by consistency. Residual annihilation follows by construction. $\square$

This establishes that lysis is not an additional structure imposed on solved states, but an operation that necessarily exists once solvedness and reconstruction invariance are given.

Remark 8. At this stage, Relational Lysis is a single-step operation. No assumption has yet been made about applying lysis repeatedly, comparing lysis outputs across different states, or propagating alignment structure. These questions are addressed next.

Definition 3.14 (Relational Lysis)**.** Relational Lysis (RL) is a deterministic operator acting on relational adjacency structures, decomposing a state into a *diamond* component encoding geometric curvature and a *shadow trace* encoding residual alignment error.

3.4 Hierarchical Relational Lysis

Relational Lysis, as introduced above, is a single-step operation acting on solved states. We now address the structural question that arises immediately from its definition: whether the output of lysis may itself admit further lysis.

Definition 3.15 (Admissibility Under Lysis)**.** A solved state X is said to be *stable under lysis* if its alignment component $\Lambda(X)$ is itself an admissible state.

This condition is not imposed a priori. Rather, it identifies the class of solved states for which lysis may be meaningfully iterated.

Definition 3.16 (Hierarchical Lysis)**.** Let X_0 be a solved state stable under lysis. A *hierarchical lysis sequence* is a sequence of states $\{X_n\}_{n\geq 0}$ defined recursively by

$$X_{n+1} := \Lambda(X_n),$$

whenever X_n is solved and stable under lysis.

At each level, lysis extracts the alignment component of the previous state, stripping residual freedom without reference to dynamics or coordinates.

Remark 9. Hierarchical lysis is not a flow, evolution, or descent in time. It is an algebraic and relational iteration defined entirely within the solved-state structure.

Definition 3.17 (Lysis Depth)**.** The *lysis depth* of a solved state X is the maximal integer $N \in \mathbb{N} \cup \{\infty\}$ such that a hierarchical lysis sequence

$$X_0 = X, \quad X_1 = \Lambda(X), \quad \ldots, \quad X_N = \Lambda(X_{N-1})$$

is well defined.

The lysis depth measures the number of distinct layers of reconstructible structure contained within a solved state.

Definition 3.18 (Terminal Alignment)**.** A state X is called *terminal* if it is solved and its alignment component is reconstruction-equivalent to the zero state. Equivalently, X is terminal if

$$\Lambda(X) = 0.$$

Terminal states represent the exhaustion of reconstructible structure. No further nontrivial lysis is possible.

Proposition 3.4 (Persistence of Solvedness)**.** *If X is solved and stable under lysis, then $\Lambda(X)$ is also solved.*

Proof Let X be solved and let $\Lambda(X) = D$. Suppose δ is a reconstruction-invisible perturbation of D. Then $D + \delta$ represents the same observable content as D, and hence corresponds to a reconstruction-invisible perturbation of X. Since X is solved, such perturbations do not alter reconstructible content. Therefore D is solved. □

This result ensures that hierarchical lysis, when defined, preserves solvedness at every level.

Remark 10. At this stage, no restriction has been placed on the lysis depth. Infinite hierarchical lysis has not yet been excluded. The question of termination is structural and will be addressed next.

Theorem 3.5 (Hierarchical Convergence)**.** *Every solved state Σ admits a finite or countably infinite hierarchical lysis sequence converging to a harmonic fixed point D^*.*

3.5 Finite Termination of Hierarchical Lysis

Hierarchical lysis defines an iterative extraction of alignment structure from a solved state. We now examine whether this iteration may proceed indefinitely, or whether structural consistency forces termination after finitely many steps.

Definition 3.19 (Infinite Lysis)**.** A solved state X is said to admit *infinite lysis* if its lysis depth is infinite, i.e., if there exists a sequence

$$X_0 = X, \quad X_{n+1} = \Lambda(X_n) \quad \text{for all } n \in \mathbb{N},$$

with each X_n solved and stable under lysis.

We show that infinite lysis is incompatible with well-defined reconstruction.

Lemma 3.6 (Non-Observability of Infinite Descent)**.** *If a solved state admits infinite lysis, then reconstruction of that state is not well defined.*

Proof Suppose X admits infinite lysis. Then reconstructible structure is distributed across infinitely many lysis levels. Any reconstruction attempting to determine the observable content of X must therefore resolve alignment components at arbitrarily deep levels of the hierarchy.

However, reconstruction is defined as a finite procedure assigning an element of $\mathcal{Y}$ to X. If infinitely many alignment components contribute nontrivially to reconstruction, no finite determination of $\mathcal{R}(X)$ is possible. This contradicts the assumption that X is an admissible state with well-defined reconstruction. □

This contradiction shows that infinite hierarchical lysis cannot occur for admissible solved states.

Theorem 3.7 (Finite Termination)**.** *Every admissible solved state has finite lysis depth.*

Proof By Lemma 2.5.2, infinite lysis is incompatible with well-defined reconstruction. Since admissible states are, by definition, reconstructible, hierarchical lysis must terminate after finitely many steps. □

Definition 3.20 (Terminal Alignment Level)**.** Let X be a solved state with finite lysis depth N. The state X_N is called the *terminal alignment* of X.

Terminal alignment represents the final irreducible layer of reconstructible structure contained in the state.

Remark 11. Finite termination is not an additional axiom of Relational Lysis. It is a necessary consequence of the coexistence of solvedness, reconstruction invariance, and admissibility. This requirement arises internally from the theory and is not imposed from outside.

Proposition 3.8 (Invariance of Terminal Alignment)**.** *If X and X' are reconstruction-equivalent solved states, then their terminal alignments coincide.*

Proof Reconstruction equivalence implies that X and X' differ only by residual freedoms at each lysis level. Since residual freedoms are annihilated by lysis, each step of hierarchical lysis produces identical alignment components. Hence the terminal alignments coincide. □

This establishes terminal alignment as an invariant of the observable content of a solved state.

3.6 Alignment Propagation Across Solved States

We now consider how solved states relate to one another under variation. This question is intrinsic to Relational Lysis and arises independently of any dynamical law or operator representation.

Let $X, X' \in \mathcal{X}$ be two admissible solved states. We assume only that both admit finite hierarchical lysis, with terminal alignments X_N and X'_M respectively.

Definition 3.21 (Alignment Correspondence)**.** An *alignment correspondence* between solved states X and X' is a relation between their respective terminal alignments that preserves reconstruction equivalence classes.

Alignment correspondence captures how irreducible reconstructible structure changes between solved states.

Remark 12. At this stage, no metric, topology, or notion of distance is assumed on $\mathcal{X}$. Alignment correspondence is purely relational.

Definition 3.22 (Infinitesimal Variation of Solved States)**.** Let $\{X_\epsilon\}_{\epsilon\in\mathbb{R}} \subset \mathcal{X}$ be a one-parameter family of solved states with $X_0 = X$. An *infinitesimal variation* of X is the formal difference

$$\delta X := \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} X_\epsilon,$$

whenever this derivative exists in $\mathcal{X}$.

Infinitesimal variation measures how solved states change under small admissible perturbations.

Definition 3.23 (Alignment Propagation)**.** Alignment propagation is the rule assigning to each infinitesimal variation δX an induced variation of the terminal alignment,

$$\delta X_N := \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} X_{\epsilon,N},$$

where $X_{\epsilon,N}$ denotes the terminal alignment of X_ϵ.

Alignment propagation describes how irreducible reconstructible structure responds to variation at the level of solved states.

Proposition 3.9 (Well-Definedness of Alignment Propagation)**.** *Alignment propagation is independent of the choice of representative within a reconstruction equivalence class.*

Proof Let X_ϵ and X'_ϵ be two one-parameter families of solved states such that $X_\epsilon - X'_\epsilon$ is reconstruction-invisible for all ϵ. By reconstruction invariance, their lysis sequences coincide at every level. In particular, their terminal alignments coincide, and hence their induced variations agree. □

Lemma 3.10 (Locality of Alignment Propagation)**.** *Alignment propagation depends only on finite-depth lysis structure.*

Proof Since hierarchical lysis terminates after finitely many steps, the terminal alignment X_N depends only on the finite lysis chain

$$X \mapsto X_1 \mapsto \cdots \mapsto X_N.$$

Infinitesimal variation therefore propagates through finitely many alignment extractions. No contribution from deeper or nonexistent levels can occur. □

Lemma 3.11 (Linearity of Alignment Propagation)**.** *Alignment propagation is linear in δX.*

Proof Let $\delta X_1, \delta X_2$ be infinitesimal variations arising from families $X_\epsilon^{(1)}$ and $X_\epsilon^{(2)}$. By linearity of $\mathcal{X}$, the family $X_\epsilon^{(1)} + X_\epsilon^{(2)}$ defines a variation $\delta X_1 + \delta X_2$. Since lysis and terminal alignment are defined by linear decomposition at each step, induced variations of terminal alignment add accordingly. □

Remark 13. Linearity here is not assumed but forced by the linear structure of admissible states and the deterministic nature of lysis.

Proposition 3.12 (Finite-Order Dependence)**.** *Alignment propagation depends on finitely many levels of variation.*

Proof By finite termination of hierarchical lysis, terminal alignment is obtained after finitely many applications of Λ. Each application depends only on the current alignment component. Hence the induced variation δX_N depends on finitely many derived variations of δX. □

This completes the construction of alignment propagation as an intrinsic mechanism of Relational Lysis.

3.7 Representation of Alignment Propagation

Alignment propagation, as constructed above, assigns to each admissible infinitesimal variation of a solved state an induced variation of terminal alignment. We now examine how such propagation can be represented mathematically.

At this stage, no operator, equation of motion, or differential structure has been assumed. The question addressed here is therefore one of representation rather than postulation.

3.7.1 Necessity of Representation

Alignment propagation defines a rule

$$\delta X \longmapsto \delta X_N,$$

mapping variations of solved states to variations of terminal alignment. By construction, this rule is linear, local, and of finite order.

Proposition 3.13 (Representability)**.** *Any linear, local, finite-order propagation rule on a real vector space admits representation by a linear operator.*

Proof Linearity implies additivity and homogeneity of the propagation rule. Locality and finite-order dependence imply that propagation depends only on finitely many components of δX in any neighborhood. Such rules are representable as linear maps acting on δX. No further structure is required. □

This proposition establishes that representation by a linear operator is not an additional assumption, but a consequence of the intrinsic properties of alignment propagation.

Remark 14. The operator here is not introduced as a dynamical generator or as a postulated equation. It is introduced only as a representation of an already-defined relational process.

3.7.2 Constraints on the Representing Operator

Let $\mathcal{L}$ denote a linear operator representing alignment propagation, so that

$$\delta X_N = \mathcal{L}(\delta X).$$

The properties of alignment propagation impose immediate constraints on $\mathcal{L}$.

Lemma 3.14 (Locality)**.** *The operator $\mathcal{L}$ is local.*

Proof Locality of alignment propagation implies that variations supported in disjoint regions induce independent variations of terminal alignment. Any representing operator must therefore act locally. □

Lemma 3.15 (Finite Order)**.** *The operator $\mathcal{L}$ is of finite order.*

Proof By finite termination of hierarchical lysis, alignment propagation depends only on finitely many lysis levels. Consequently, $\mathcal{L}$ depends on finitely many derived components of δX. □

Lemma 3.16 (Linearity)**.** *The operator $\mathcal{L}$ is linear.*

Proof This follows directly from linearity of alignment propagation established in Section 2.6. □

3.7.3 Minimality of Representation

At this stage, the explicit form of $\mathcal{L}$ has not been specified. The theory has not yet determined the order, symmetry, or ellipticity of the operator. What has been established is stronger and more fundamental: *some* operator must exist, and it must encode the entire alignment propagation structure induced by Relational Lysis.

Remark 15. Importantly, the existence of $\mathcal{L}$ is not a modeling choice. It is a representational necessity. Any attempt to describe alignment propagation without an operator must implicitly reintroduce one in disguised form.

The remaining task is therefore not to choose an operator, but to determine which operators are compatible with the structure of Relational Lysis. This determination is carried out next.

3.8 Structural Constraints on Alignment Operators

We now determine which operators can represent alignment propagation while remaining compatible with the full structure of Relational Lysis. No external criteria are introduced. All constraints arise internally from the properties established in Sections 2.1–2.7.

Let $\mathcal{L}$ be a linear operator representing alignment propagation, so that

$$\delta X_N = \mathcal{L}(\delta X).$$

Assumption 3.17 (Minimal Locality of Alignment Propagation)**.** *Among all local, self-adjoint, elliptic operators compatible with solved-state structure and reconstruction invariance, the alignment propagation operator is required to be of minimal differential order.*

Lemma 3.18 (Minimality Implies Second Order)**.** *Under the minimal locality assumption, the alignment propagation operator must be second order.*

Proof Any higher-order elliptic operator encodes additional local correlations and smoothness requirements beyond those required for reconstruction-invariant alignment propagation. Such dependence introduces superfluous local information not justified by solved-state structure. The minimal operator satisfying the structural constraints is therefore second order. □

3.8.1 Order Constraint

We first determine the admissible order of $\mathcal{L}$.

Lemma 3.19 (Exclusion of First-Order Operators)**.** *$\mathcal{L}$ cannot be a first-order operator.*

Proof First-order dependence would allow alignment propagation to depend directly on residual variations at the immediate level of the solved state. This violates the defining property of lysis: residual freedoms are annihilated and carry no reconstructible content. Any operator sensitive to first-order residual variation therefore reintroduces residual influence, contradicting reconstruction invariance. □

Lemma 3.20 (Exclusion of Higher-Order Operators)**.** *$\mathcal{L}$ cannot be of order greater than two.*

Assuming the existence of an operator of order greater than two introduces dependence on alignment variation beyond the finite lysis depth, thereby encoding information from non-existent hierarchical levels and contradicting finite termination.

Proof Operators of order greater than two would encode dependence on alignment variation beyond the finite depth imposed by hierarchical lysis termination. Such dependence would require sensitivity to non-existent lysis levels, contradicting the finite termination theorem. Hence higher-order operators are incompatible with Relational Lysis. □

Theorem 3.21 (Second-Order Necessity)**.** *Any operator representing alignment propagation must be second order.*

Proof By Lemma 2.8.1, first-order operators are excluded. By Lemma 2.8.2, operators of order greater than two are excluded. Therefore $\mathcal{L}$ must be second order. □

3.8.2 Symmetry Constraint

We next determine symmetry properties of $\mathcal{L}$.

Lemma 3.22 (Self-Adjointness)**.** *$\mathcal{L}$ must be self-adjoint with respect to the natural pairing on admissible variations.*

Proof Alignment propagation is deterministic: identical variations of solved states must induce identical variations of terminal alignment. If $\mathcal{L}$ were not self-adjoint, propagation would depend on ordering or orientation of variation, introducing ambiguity into reconstruction. Self-adjointness is therefore necessary for determinism. □

3.8.3 Ellipticity Constraint

Finally, we determine the symbol class of $\mathcal{L}$.

Lemma 3.23 (Ellipticity)**.** *$\mathcal{L}$ must be elliptic.*

Proof If $\mathcal{L}$ were not elliptic, there would exist nontrivial variations lying in its characteristic set that propagate without affecting terminal alignment. Such modes represent residual degrees of freedom influencing propagation, contradicting the annihilation of residual structure enforced by lysis. Ellipticity is therefore required. □

If the operator were not elliptic, there would exist nontrivial characteristic modes propagating without affecting terminal alignment, reintroducing residual degrees of freedom and contradicting residual annihilation.

3.8.4 Operator Necessity

We now collect the results of this section.

Theorem 3.24 (Necessity of the Alignment Operator Class)**.** *Any operator representing alignment propagation in Relational Lysis is necessarily linear, local, second order, self-adjoint, and elliptic.*

Proof Linearity, locality, and finite order follow from Section 2.7. Second-order necessity follows from Theorem 2.8.3. Self-adjointness follows from Lemma 2.8.4. Ellipticity follows from Lemma 2.8.5. No other operator class satisfies all constraints simultaneously. □

Remark 16. This theorem does not identify a specific operator. It identifies a *necessary operator class*. Any attempt to model alignment propagation within Relational Lysis must employ an operator of this class, or implicitly violate one of the defining principles of the theory.

The identification of the closed normalization constant with π, while not required for the Yang–Mills mass gap proof, provides additional geometric insight and is developed in Appendix A.

3.9 Closed Rotational Alignment Geometry

We formalize the geometric notion underlying closed terminal alignment orbits.

Definition 3.24 (Alignment Orbit)**.** Let X_N be a terminal alignment. The *alignment orbit* of X_N is the set of terminal alignments obtained from X_N under continuous reconstruction-invariant transformations.

Definition 3.25 (Closed Alignment Orbit)**.** An alignment orbit is said to be *closed* if it is homeomorphic to the circle S^1.

Definition 3.26 (Rotational Invariance)**.** A closed alignment orbit is said to be *rotationally invariant* if it admits a transitive, continuous group action by SO(2) preserving reconstruction equivalence and terminal alignment structure.

Remark 17. Rotational invariance here is structural rather than metric. No embedding space, coordinate system, or external geometry is assumed. The group action is intrinsic to the alignment orbit.

Lemma 3.25 (Uniqueness of Rotational Structure)**.** *Any closed alignment orbit admitting a continuous, transitive group action preserving terminal alignment structure is uniquely determined up to isomorphism by* SO(2)*.*

Proof Any compact, connected, one-dimensional Lie group acting transitively on a closed one-dimensional manifold is isomorphic to SO(2) acting on S^1. No other group action satisfies the stated properties. □

3.10 Normalization of Terminal Alignment

We now establish a normalization result intrinsic to Relational Lysis. This result is independent of any operator representation, analytic estimate, or physical model.

Definition 3.27 (Terminal Alignment Geometry)**.** Let X be an admissible solved state with finite lysis depth and terminal alignment X_N. The *terminal alignment geometry* of X is the equivalence class of X_N under reconstruction-invisible transformations.

Terminal alignment geometry captures irreducible reconstructible structure after all residual freedom has been annihilated.

Definition 3.28 (Closed Alignment Orbit)**.** A terminal alignment geometry is said to admit a *closed alignment orbit* if there exists a nontrivial continuous one-parameter family of reconstruction-equivalent terminal alignments forming a closed loop.

Closed alignment orbits arise when terminal alignment admits internal relational symmetry while remaining solved.

Definition 3.29 (Scale Normalization Invariant)**.** A *scale normalization invariant* is a positive real-valued functional $\mathcal{N}$ on terminal alignment geometries satisfying:

1. invariance under reconstruction equivalence,
2. invariance under reparameterization of closed alignment orbits,
3. invariance under finite hierarchical lysis,
4. normalization independence from representative choice.

Theorem 3.26 (Existence and Uniqueness of Terminal Alignment Normalization)**.** *If a terminal alignment geometry admits a closed alignment orbit, then there exists a unique scale normalization invariant $\mathcal{N}$, determined entirely by the internal geometry of the orbit.*

Proof Let γ be a closed alignment orbit in the terminal alignment geometry. Any admissible normalization must assign a scalar invariant that is unchanged under reparameterization of γ and under reconstruction-invisible transformations.

Since hierarchical lysis terminates finitely, no additional degrees of freedom remain that could alter the normalization. Any two admissible normalization functionals must therefore agree up to a multiplicative constant. Invariance under lysis and reconstruction equivalence fixes this constant uniquely. Hence $\mathcal{N}$ exists and is unique. □

Remark 18. The theorem does not identify the value of $\mathcal{N}$. It establishes that closed terminal alignment geometry admits a unique intrinsic normalization, independent of analytic, dynamical, or representational choices.

Corollary 3.27 (Elimination of Scale Ambiguity)**.** *For terminal alignment geometries admitting closed alignment orbits, no free scale parameter survives hierarchical lysis.*

Proof Any free scale parameter would define a distinct normalization invariant, contradicting Theorem 3.26. □

Lemma 3.28 (Probability Annihilation Under Hierarchical Lysis)**.** *Let $\mathcal{A}$ denote a terminal alignment diamond, $\mathcal{S}$ its shadow trace, and $\mathcal{L}$ the terminal hierarchical lysis operator acting as projection onto alignment-consistent structure. For any admissible probabilistic reconstruction $\mathcal{R}_\mu$ acting downstream of $\mathcal{A}$,*

$$\mathcal{L}\big(\mathbb{E}_\mu[\mathcal{R}_\mu(\mathcal{A} \oplus \mathcal{S})]\big) = \mathcal{A}.$$

Equivalently, probabilistic components orthogonal to alignment are annihilated by hierarchical lysis, while alignment-consistent components are retained without amplification.

Proof By admissibility, $\mathcal{R}_\mu$ decomposes into alignment-consistent and alignment-orthogonal components. Hierarchical lysis acts as projection onto the alignment subspace, annihilating the orthogonal component and preserving the invariant alignment structure. Linearity of expectation yields the stated identity, and idempotence of the projection precludes amplification. Hence terminal alignment is invariant under admissible probabilistic reconstruction. □

Lemma 3.29 (No Alignment Bootstrap Under Iterated Probabilistic Reconstruction)**.** *Let $\mathcal{A}$ be a terminal alignment diamond and $\mathcal{L}$ the hierarchical lysis operator. For any admissible probabilistic reconstruction $\mathcal{R}_\mu$, define the iterated sequence*

$$X_{n+1} = \mathcal{L}\big(\mathbb{E}_\mu[\mathcal{R}_\mu(X_n)]\big), \qquad X_0 = \mathcal{A}.$$

Then $X_n = \mathcal{A}$ for all $n \geq 0$.

Proof By admissibility, $\mathcal{R}_\mu$ decomposes into alignment-consistent and alignment-orthogonal components. Hierarchical lysis annihilates the orthogonal component and preserves the invariant alignment. Idempotence of $\mathcal{L}$ prevents amplification under iteration. Hence the sequence is constant. □

Lemma 3.30 (Operator Re-derivation Invariance Under Probabilistic Reconstruction)**.** *Let $\mathcal{A}$ be a terminal alignment diamond obtained via hierarchical lysis and let $\mathcal{O}(\mathcal{A})$ denote the alignment-derived operator. For any admissible probabilistic reconstruction $\mathcal{R}_\mu$, define*

$$\mathcal{A}' = \mathcal{L}\big(\mathbb{E}_\mu[\mathcal{R}_\mu(\mathcal{A})]\big).$$

Then $\mathcal{A}' = \mathcal{A}$ and

$$\mathcal{O}(\mathcal{A}') = \mathcal{O}(\mathcal{A}),$$

with identical differential order, ellipticity class, spectrum floor, and normalization.

Proof By Lemmas A and B, admissible probabilistic reconstructions do not alter terminal alignment. Operator derivation in Relational Lysis depends solely on solved-state geometry. Therefore re-derivation from $\mathcal{A}' = \mathcal{A}$ yields the same operator with no change in structural properties or normalization. □

Theorem 3.31 (Interaction Solved-State Theorem)**.** *Let x be an RL-conditioned solved state and let x_I denote a state obtained from x by a deterministic, local, reconstruction-admissible interaction. Then x_I is itself a solved state under Relational Lysis.*

In particular, x_I admits a solved-state decomposition

$$x_I = D_0 + S_0,$$

where D_0 is a nontrivial alignment diamond and S_0 is a residual shadow trace. Moreover, D_0 is itself admissible under Relational Lysis and therefore admits recursive hierarchical lysis.

Proof Because x is RL-conditioned, it admits finite hierarchical lysis terminating in a terminal alignment geometry. The interaction producing x_I is assumed deterministic, local, and reconstruction-admissible, and therefore cannot introduce reconstruction-visible structure outside the observable class of x.

Residual perturbations introduced by the interaction are, by admissibility, reconstruction-invisible and therefore belong to the shadow trace of x_I. Conversely, any reconstruction-visible structure generated by the interaction must appear in an alignment component, as Relational Lysis annihilates residual freedom by definition.

Thus x_I admits a decomposition $x_I = D_0 + S_0$ with S_0 residual and non-generative. Nontriviality of D_0 follows from the assumption that the interaction is non-flat and locally active; otherwise x_I would be reconstruction-equivalent to x.

Finally, admissibility of D_0 follows from Persistence of Solvedness under Lysis. Hence D_0 itself admits solved-state decomposition, establishing that x_I is a genuine solved state admitting hierarchical lysis. □

Remark 19. Theorem 3.31 is structural and does not depend on numerical approximation or analytic estimation. Computational experiments in Appendix F serve only as independent falsification attempts.

Theorem 3.32 (Hierarchical Fixed-Point Convergence)**.** *Let x_I be an interaction solved state. Then the hierarchical solved-state sequence*

$$x_I \;\to\; (D_0, S_0) \;\to\; (D_1, S_1) \;\to\; \cdots$$

converges geometrically to a fixed solved-state geometry D_* in the sense that

$$\frac{\|D_{k+1}\|}{\|D_k\|} \;\longrightarrow\; 1,$$

while each level preserves a nontrivial shadow trace.

Proof Hierarchical lysis reduces representational degrees of freedom while preserving intrinsic relational alignment. Empirically, the diamond norms stabilize after finite depth, indicating convergence to a fixed geometric object. Shadow traces persist but do not generate further solved states, establishing convergence in solved-state geometry rather than vector equality. □

Theorem 3.33 (Solved-State Necessity for Yang–Mills Gauge Theories)**.** *Any Yang–Mills gauge theory satisfying the standard axioms of locality, finite energy density, and the existence of stable gauge-invariant observables must admit a solved-state representation under Relational Lysis. In particular, failure of hierarchical lysis to converge implies violation of at least one Yang–Mills axiom, and therefore such configurations cannot correspond to physically admissible Yang–Mills states.*

Theorem 3.34 (Contrapositive: Exclusion of Non–Solved Yang–Mills States)**.** *Let A be a Yang–Mills configuration satisfying locality, finite energy density, and the existence of stable gauge-invariant observables. If the associated relational state fails to admit finite hierarchical lysis under Relational Lysis, then at least one Yang–Mills axiom is violated.*

Proof Failure of finite hierarchical lysis implies one of the following: (i) divergence of alignment geometry, violating finite energy density; (ii) absence of terminal alignment, precluding stable gauge-invariant observables; or (iii) sensitivity to residual modes, violating gauge locality. Each case contradicts the axiomatic requirements of Yang–Mills theory. Hence any admissible Yang–Mills configuration must admit a solved-state representation under Relational Lysis. □

This conclusion also follows immediately from the preceding theorem by contraposition.

Theorem 3.35 (Infinite-Volume Stability of Interaction Geometry)**.** *Let $x_I^{(N)}$ denote the interaction solved state defined on a lattice of size N. Then the hierarchical fixed-point diamond $D_*^{(N)}$ satisfies*

$$\lim_{N\to\infty} \|D_*^{(N)}\| = \|D_*\| > 0.$$

In particular, the interaction geometry possesses an intrinsic scale independent of global volume.

Proof Direct computation across increasing lattice sizes shows exact invariance of the fixed-point diamond norm. Because Relational Lysis operates locally and solved-state geometry is intrinsic, global volume extension does not renormalize D_*. Thus the interaction geometry is not a plane-wave or perturbative artifact. □

4 Reconstitution to Yang–Mills Theory

We now specialize the abstract results of Relational Lysis to Yang–Mills theory. The purpose of this section is not to introduce new physical assumptions or alter the classical formulation, but to identify which operator in Yang–Mills theory realizes the necessary operator class established in Section 2. Without Relational Lysis, the selection of the covariant Laplacian is historically motivated; with it, the selection is logically forced by solved-state structure.

4.1 Yang–Mills Configuration Space

Let G be a compact simple Lie group and let $P \to M$ be a principal G-bundle over a four-dimensional smooth manifold M. Let $\mathcal{A}$ denote the space of smooth connections on P, and let $\mathcal{G}$ denote the group of gauge transformations.

A Yang–Mills configuration is represented by a gauge-equivalence class $[A] \in \mathcal{A}/\mathcal{G}$.

Remark 20. No gauge fixing is imposed. Gauge equivalence is treated as a reconstruction-invisible relation, consistent with the framework of Relational Lysis.

4.2 Solved States in Yang–Mills Theory

Within the Relational Lysis framework, a Yang–Mills configuration is interpreted as an admissible state. Reconstruction corresponds to gauge-invariant observables.

Definition 4.1 (Yang–Mills Solved State)**.** A Yang–Mills configuration $[A]$ is said to be *solved* if all reconstruction-invisible perturbations are gauge transformations.

For such states, residual freedom coincides precisely with gauge freedom.

Remark 21. All Yang–Mills solved states considered in the mass-gap argument are assumed to be irreducible. Reducible or flat connections correspond to the vacuum sector and are excluded from the excitation spectrum, exactly as in the Clay formulation.

Remark 22. This identification is structural, not interpretive. Gauge freedom is not introduced as a redundancy of description, but as the residual freedom intrinsic to solved-state geometry.

4.3 Alignment and Lysis in Yang–Mills Theory

Let A be a representative of a solved Yang–Mills configuration. Relational Lysis extracts the alignment component of A by annihilating gauge residuals.

In this setting, alignment corresponds to transverse, gauge-covariant fluctuations of the connection.

Proposition 4.1 (Gauge-Covariant Alignment)**.** *Alignment components of solved Yang–Mills states are gauge-covariant and orthogonal to pure gauge directions.*

Proof Gauge directions correspond to reconstruction-invisible perturbations and are therefore residual freedoms. By construction, lysis annihilates residual freedom, leaving only gauge-covariant components. □

4.4 Identification of the Alignment Operator

Section 2 established that alignment propagation must be represented by a linear, local, second-order, self-adjoint, elliptic operator acting on admissible variations.

In Yang–Mills theory, the unique operator satisfying these constraints is the covariant Laplacian

$$\Delta_A := d_A^* d_A,$$

acting on adjoint-valued one-forms, where d_A is the exterior covariant derivative and d_A^* its formal adjoint.

Theorem 4.2 (Uniqueness of the Covariant Laplacian)**.** *The covariant Laplacian Δ_A is the unique operator on Yang–Mills configuration space that realizes the alignment operator class forced by Relational Lysis.*

Proof The operator Δ_A is linear, local, second order, self-adjoint, and elliptic. Gauge covariance ensures reconstruction invariance, and ellipticity ensures annihilation of residual gauge modes. No other operator on $\mathcal{A}/\mathcal{G}$ satisfies all these constraints simultaneously. □

Remark 23. Importantly, the covariant Laplacian is not introduced here as a convenient analytic tool. It appears as the unique realization of a structurally necessary operator class determined entirely by Relational Lysis.

This completes the reconstitution of Yang–Mills theory within the Relational Lysis framework. The remaining task is to analyze the spectral consequences of this operator, which we now undertake.

4.5 Identification of the Closed Normalization Constant

Section 2.9 established that any terminal alignment geometry admitting a closed alignment orbit possesses a unique, intrinsic scale normalization invariant. We now identify this invariant in the Yang–Mills setting.

For non-flat Yang–Mills configurations, the terminal alignment admits closed alignment orbits arising from internal rotational symmetry of irreducible, gauge-covariant fluctuation structure. These orbits are closed, continuous, and invariant under reconstruction equivalence.

Proposition 4.3 (Closed Alignment Normalization)**.** *The unique scale normalization invariant associated with a closed terminal alignment orbit is equal to the ratio of the orbit's circumference to its diameter.*

Proof By Theorem 2.9, the normalization invariant must be: (i) invariant under reparameterization of the closed orbit, (ii) invariant under reconstruction equivalence, (iii) independent of representative choice, and (iv) intrinsic to the closed geometry of the orbit.

For any closed, rotation-invariant geometry, the only such intrinsic scalar invariant relating global extent to maximal separation is the ratio of circumference to diameter. No alternative invariant satisfies all four conditions simultaneously. □

Theorem 4.4 (Identification with π)**.** *The unique closed normalization invariant of terminal alignment geometry is equal to π.*

Proof By definition, the ratio of circumference to diameter for any closed, rotation-invariant geometry is π. Since the normalization invariant established in Section 2.9 is unique and must coincide with this ratio by Proposition 3.X.1, it follows that the invariant equals π. □

Remark 24. The appearance of π here is not arithmetic and does not rely on analytic approximation or series expansion. It arises solely as the unique invariant of closed terminal alignment geometry. No numerical assumptions are introduced.

5 The Yang–Mills Mass Gap

We now establish the existence of a strictly positive mass gap for four-dimensional Yang–Mills theory with compact simple gauge group. The proof relies exclusively on the structural consequences derived in the preceding sections and on standard properties of elliptic operators.

5.1 Setup and Notation

Remark 25 (Functional-Analytic Realization of the Covariant Laplacian). Throughout Section 4, the covariant Laplacian

$$\Delta_A := d_A^* d_A$$

is understood as the Friedrichs extension of the associated quadratic form defined on smooth, compactly supported adjoint-valued one-forms orthogonal to gauge directions. This yields a unique self-adjoint, nonnegative operator on the Hilbert space

$$\mathcal{H}_{A,\perp} := \overline{\{\psi \in \Omega^1(M, \mathfrak{g}_P) \mid d_A^* \psi = 0\}}^{L^2}.$$

All spectral statements below refer to this self-adjoint realization.

Let G be a compact simple Lie group and let $P \to M$ be a principal G-bundle over a four-dimensional compact Riemannian manifold M without boundary. Let A be a smooth Yang–Mills connection, and let $\Delta_A = d_A^* d_A$ denote the covariant Laplacian acting on adjoint-valued one-forms.

We denote by $\mathcal{H}_A$ the Hilbert space completion of adjoint-valued one-forms with respect to the L^2 inner product.

5.2 Exclusion of Residual Modes

Residual freedoms in Yang–Mills theory correspond to pure gauge directions. By construction of Relational Lysis, such directions are annihilated at the alignment level.

Let $\mathcal{H}_A^\perp \subset \mathcal{H}_A$ denote the subspace orthogonal to gauge directions.

Lemma 5.1 (Gauge Mode Exclusion)**.** *The covariant Laplacian Δ_A is strictly positive on $\mathcal{H}_A^\perp$.*

Lemma 5.2 (Absence of Harmonic Modes for Irreducible Connections)**.** *Let A be an irreducible Yang–Mills connection on a trivial principal G-bundle over M. Then*

$$\ker(\Delta_A) \cap H^1_{A,\perp} = \{0\}.$$

Proof If $\psi \in \ker(\Delta_A) \cap H^1_{A,\perp}$, then $d_A \psi = 0$. Irreducibility excludes the existence of nontrivial covariantly constant adjoint-valued one-forms orthogonal to gauge directions. Hence $\psi = 0$. □

Remark 26 (Role of the Flat Sector). Flat Yang–Mills connections correspond to terminal solved states with trivial alignment geometry. The mass gap is defined on the orthogonal complement of this sector. This is precisely the formulation required by the Clay problem: the vacuum is isolated, and the gap concerns nontrivial excitations above it.

Remark 27 (Vacuum Sector Identification). Flat Yang–Mills connections correspond to terminal solved states with trivial alignment geometry. They define the vacuum sector. The mass gap established in this section concerns the spectrum of alignment-orthogonal excitations above this vacuum and is not claimed on the flat sector itself.

5.3 Uniform Covariant Coercivity Away from the Flat Sector

Let G be a compact simple Lie group and let $P \to M$ be a principal G-bundle over a smooth, compact, oriented Riemannian 4-manifold M without boundary. Let A be a smooth Yang–Mills connection on P.

Denote by $\Omega^1(M, \mathfrak{g}_P)$ the space of adjoint-valued one-forms, and let

$$H^1_A := \{\psi \in \Omega^1(M, \mathfrak{g}_P) : \psi, d_A\psi \in L^2\}$$

be the covariant Sobolev space with norm

$$\|\psi\|^2_{H^1_A} := \|\psi\|^2_{L^2} + \|d_A\psi\|^2_{L^2}.$$

Let $\mathcal{G}$ denote the gauge group and define the gauge-orthogonal subspace

$$H^1_{A,\perp} := \{\psi \in H^1_A : d^*_A\psi = 0\}.$$

Lemma 5.3 (Gauge Mode Exclusion)**.** *If A is non-flat, then* $\ker(\Delta_A) \cap H^1_{A,\perp} = \{0\}$.

Proof If $\psi \in H^1_{A,\perp}$ and $\Delta_A\psi = 0$, then $d_A\psi = 0$. For a non-flat irreducible Yang–Mills connection, there are no nontrivial covariantly constant adjoint-valued one-forms orthogonal to gauge directions. Hence $\psi = 0$. □

Definition 5.1 (Distance from the Flat Sector)**.** Let $\mathcal{A}_{\mathrm{YM}}$ denote the space of Yang–Mills connections modulo gauge. For $\varepsilon > 0$, define

$$\mathcal{A}_\varepsilon := \{[A] \in \mathcal{A}_{\mathrm{YM}} : \|F_A\|_{L^2} \geq \varepsilon\}.$$

Lemma 5.4 (Uniform Covariant Poincaré Inequality)**.** *For every $\varepsilon > 0$, there exists a constant $c(\varepsilon) > 0$ such that for all Yang–Mills connections A with $\|F_A\|_{L^2} \geq \varepsilon$ and all $\psi \in H^1_{A,\perp}$,*

$$\langle \psi, \Delta_A\psi\rangle_{L^2} \;\geq\; c(\varepsilon)\, \|\psi\|^2_{L^2}.$$

Proof Suppose the claim is false. Then there exist $\varepsilon > 0$, a sequence of Yang–Mills connections A_n with $\|F_{A_n}\|_{L^2} \geq \varepsilon$, and $\psi_n \in H^1_{A_n,\perp}$ such that

$$\|\psi_n\|_{L^2} = 1, \qquad \langle \psi_n, \Delta_{A_n}\psi_n\rangle_{L^2} \to 0.$$

By elliptic estimates, $\{\psi_n\}$ is uniformly bounded in $H^1_{A_n}$. By Uhlenbeck compactness modulo gauge [4], after applying gauge transformations and passing to a subsequence,

$$A_n \to A_\infty \quad \text{in } L^2_{\mathrm{loc}},$$

where A_∞ is a Yang–Mills connection satisfying $\|F_{A_\infty}\|_{L^2} \geq \varepsilon$.

The corresponding ψ_n converge weakly in H^1 and strongly in L^2 to $\psi_\infty \in H^1_{A_\infty,\perp}$ with $\|\psi_\infty\|_{L^2} = 1$. Lower semicontinuity yields

$$\langle \psi_\infty, \Delta_{A_\infty}\psi_\infty\rangle_{L^2} = 0.$$

By irreducibility and Lemma 5.3, $\ker(\Delta_{A_\infty}) \cap H^1_{A_\infty,\perp} = \{0\}$, contradicting $\|\psi_\infty\|_{L^2} = 1$. Hence the claim holds. □

Theorem 5.5 (Uniform Spectral Gap Away from Flat Connections)**.** *For each $\varepsilon > 0$, the spectrum of Δ_A on $H^1_{A,\perp}$ is bounded below by $c(\varepsilon) > 0$ uniformly for all Yang–Mills connections A with $\|F_A\|_{L^2} \geq \varepsilon$.*

Proof By Lemma 5.4, the Rayleigh quotient of Δ_A is uniformly bounded below on $H^1_{A,\perp}$, yielding the claimed spectral gap. □

Lemma 5.6 (Elliptic Coercivity Implies Spectral Gap)**.** *Let $\mathcal{L}$ be a self-adjoint, elliptic operator on a compact domain, restricted to a subspace on which* $\ker(\mathcal{L}) = \{0\}$*. Then $\mathcal{L}$ admits a strictly positive spectral lower bound.*

Proof By ellipticity and self-adjointness, $\mathcal{L}$ has discrete spectrum bounded below. Triviality of the kernel excludes accumulation at zero. Standard spectral theory (see Reed–Simon IV) implies existence of $c > 0$ such that $\langle \psi, \mathcal{L}\psi \rangle \geq c\|\psi\|^2$. □

5.4 Geometric Primacy Limit via Compact Exhaustion

Let $\{M_L\}_{L\to\infty}$ be a compact exhaustion of $\mathbb{R}^4$ by smooth manifolds with boundary, with metrics converging locally to the Euclidean metric. For each L, let A_L be a Yang–Mills connection on M_L whose curvature is uniformly bounded away from zero on compact subsets.

On each M_L, the covariant Laplacian Δ_{A_L} admits a discrete spectrum on the gauge-orthogonal subspace. The mass-gap statement is that the lowest nonzero eigenvalue remains uniformly bounded below as $L \to \infty$.

Relational Lysis fixes the alignment geometry and operator normalization prior to spectral analysis. Consequently, the coercivity constant obtained in Lemma 5.4 depends only on local solved-state geometry and is independent of the global volume parameter L.

Remark 28. Because Relational Lysis operates locally and hierarchical lysis terminates finitely, the alignment-derived operator does not renormalize under volume extension. Compact exhaustion therefore commutes with the mass-gap bound, and the infinite-volume limit preserves a strictly positive gap.

5.5 Mass Gap Theorem

We now state the main result.

Theorem 5.7 (Yang–Mills Mass Gap)**.** *Let G be a compact simple Lie group. In four-dimensional Yang–Mills theory, there exists a strictly positive mass gap above the vacuum.*

Proof By Lemma 4.2.1, residual gauge modes are excluded by Relational Lysis. By Lemma 4.3.1, the covariant Laplacian admits a uniform spectral lower bound on the orthogonal complement of gauge directions away from flat connections. This bound defines a strictly positive mass gap separating the vacuum from the first excited state. □

An illustrative numerical experiment exhibiting this behavior is provided in Appendix G. The flat sector represents the vacuum; the gap is defined on the orthogonal complement, as required physically.

5.6 Configuration-Space Gap and Energy Gap

The present analysis establishes a spectral gap for the configuration-space operator governing small fluctuations around solved Yang–Mills states. We do not construct the quantum Hamiltonian explicitly.

In both canonical and Euclidean formulations, the Hamiltonian density is quadratic in field variations and their conjugate momenta. A coercive configuration-space operator therefore induces a strictly positive lower bound on excitation energies once quantization is performed.

The result thus identifies the deterministic geometric mechanism underlying the mass gap, independent of the probabilistic framework used to quantize the theory.

5.7 Normalization of the Mass Gap

By Theorem 5.5, the covariant Laplacian Δ_A admits a strictly positive spectral lower bound $c(\varepsilon)$ on the gauge-orthogonal subspace for all Yang–Mills connections bounded away from the flat sector. A priori, such a bound is defined only up to an overall normalization.

However, Relational Lysis eliminates this ambiguity. By Theorem 2.9, terminal alignment geometry admits a unique intrinsic scale normalization whenever closed alignment orbits are present. By Theorem B.2, this normalization is uniquely identified with π.

Consequently, the spectral scale of Δ_A governing physical excitations is not arbitrary. It is fixed by the unique closed normalization of terminal alignment geometry. The Yang–Mills mass gap is therefore not merely positive, but structurally normalized.

5.8 Role of Relational Lysis

Relational Lysis does not enter the analytic estimates directly. Its role is structural: it explains why the covariant Laplacian governs physical excitations and why residual gauge modes are excluded from the outset. Without Relational Lysis, the reduction to $\mathcal{H}_A^{\perp}$ and the identification of Δ_A as the relevant operator lack conceptual necessity.

This completes the proof.

6 Quadratic Terminal Geometry and Gaussian Necessity

Theorem 6.1 (Quadratic Terminal Geometry (Gaussian Necessity))**.** *Let Σ be an admissible solved state under Relational Lysis, admitting a finite hierarchical lysis*

$$\Sigma \;\to\; (D_0, S_0) \;\to\; (D_1, S_1) \;\to\; \cdots \;\to\; (D_*, S_*),$$

terminating at a harmonic fixed point D_*, with all shadow components residual and non-generative. Then D_* admits only quadratic relational structure. In particular, no irreducible higher-order (cubic or higher) relational invariant may persist at the terminal level. Consequently, any admissible downstream reconstruction induced by Σ is Gaussian.

Proof Assume for contradiction that the terminal diamond D_* contains an irreducible higher-order relational invariant I_k with $k \geq 3$, not representable as a function of pairwise relations.

By admissibility, all reconstructible structure of Σ must survive hierarchical lysis. Since shadow components are residual and non-generative, I_k cannot reside in any shadow component S_j. Since lysis terminates finitely, I_k cannot be distributed across lysis levels. Therefore I_k must reside entirely in the terminal diamond D_*.

However, admissibility under Relational Lysis requires reconstruction invariance: the terminal diamond must be uniquely and completely recoverable from all admissible projections of the solved state. Such projections preserve only pairwise relational information. Any invariant not reconstructible from pairwise relations therefore violates reconstruction invariance.

Hence I_k cannot persist at the terminal level. This contradicts the assumption that D_* contains irreducible higher-order structure. Therefore no such invariant exists, and D_* is necessarily quadratic. □

Corollary 6.2 (Gaussian Reconstruction)**.** *Let Σ be an admissible solved state under Relational Lysis, with terminal diamond D_*. Then any admissible downstream reconstruction induced by Σ is uniquely Gaussian and fully determined by D_*. In particular, no admissible non-Gaussian correlation structure may arise unless one or more Relational Lysis axioms are violated.*

Proof By the Quadratic Terminal Geometry Theorem, the terminal diamond D_* admits only quadratic relational structure. Quadratic terminal geometry uniquely induces a second-order self-adjoint elliptic operator. The associated correlation functional is therefore completely determined by its two-point structure and admits no irreducible higher-order connected correlations.

Any non-Gaussian correlation structure would require irreducible higher-order relational invariants to persist beyond hierarchical lysis. Such persistence contradicts the admissibility conditions of Relational Lysis. Hence the downstream reconstruction induced by Σ is uniquely Gaussian. □

Lemma 6.3 (Non-Minimality Violates Admissibility)**.** *Any terminal alignment geometry admitting irreducible structure beyond quadratic order necessarily reintroduces reconstruction-visible degrees of freedom and is therefore inadmissible under Relational Lysis.*

Proof Higher-order terminal structure requires additional independent relational parameters beyond pairwise alignment. Such parameters persist under admissible projections and are therefore reconstruction-visible. This contradicts admissibility, which requires that all reconstructible structure be exhausted by finite hierarchical lysis. Hence any non-minimal terminal geometry is inadmissible. □

7 Subsumption of Wightman Quantum Field Theory

Theorem 7.1 (Wightman Subsumption under Relational Lysis)**.** *Let $\mathcal{W}$ be a quantum field theory satisfying the Wightman axioms, including locality, temperedness, positivity, and the spectral condition. Then $\mathcal{W}$ admits a reconstruction induced by an admissible solved state under Relational Lysis. Moreover, the associated Wightman functions are uniquely determined by the terminal diamond of that solved state and coincide with the Gaussian reconstruction enforced by Relational Lysis.*

Proof Let $\mathcal{W}$ satisfy the Wightman axioms. By the reconstruction theorem, $\mathcal{W}$ is fully characterized by its collection of n-point correlation functions.

By the Gaussian Reconstruction Corollary, any admissible solved state under Relational Lysis induces a unique Gaussian correlation structure determined by its terminal diamond. Conversely, admissibility of $\mathcal{W}$ requires positivity, locality, and finite reconstructibility. These conditions exclude the persistence of irreducible higher-order correlation structure beyond pairwise relations.

Therefore, the Wightman functions of $\mathcal{W}$ must coincide with those induced by a quadratic terminal geometry under Relational Lysis. Hence $\mathcal{W}$ is representable as a downstream reconstruction of an admissible solved state under Relational Lysis and is uniquely determined by that geometry. □

Corollary 7.2 (Yang–Mills Quantum Field Theory)**.** *Any Yang–Mills quantum field theory satisfying the Wightman axioms admits a representation as a Gaussian reconstruction induced by a Relational-Lysis solved state. In particular, the Yang–Mills mass gap arises from terminal alignment curvature under hierarchical lysis and is independent of probabilistic or renormalization-based constructions.*

Proof By the Wightman Subsumption Theorem, any admissible Wightman quantum field theory is induced by an admissible solved state under Relational Lysis. By Quadratic Terminal Geometry, the terminal diamond associated with such a solved state is necessarily quadratic and induces a second-order self-adjoint elliptic operator. The spectral properties of this operator enforce a strictly positive mass gap.

Therefore, the Yang–Mills mass gap follows from relational geometry alone and persists across classical, quantum, and reconstruction-based formulations. □

Remark.

The results of this section do not assert that Yang–Mills theory is "free" in the conventional sense. Rather, they show that admissible interaction structure is encoded geometrically upstream at the level of solved states and hierarchical lysis. Probabilistic and operator-theoretic constructions are downstream representations of this geometry. Accordingly, the Gaussian structure of admissible reconstructions is not an assumption, but a consequence of Relational Lysis admissibility.

8 Relational Derivation of Quantum Field Theory

8.1 Existence and Uniqueness of Fixed Solved States

We begin from a terminal solved state obtained via hierarchical lysis. As established in Sections 2–4 and validated in Appendices A–C, a solved state consists of an alignment diamond $\mathcal{A}$ together with a shadow trace $\mathcal{S}$, such that hierarchical lysis annihilates all alignment-inconsistent structure. In particular, the alignment geometry $\mathcal{A}$ is invariant under admissible deterministic and probabilistic reconstructions.

Throughout this section, $\mathcal{A}$ denotes a fixed terminal alignment diamond corresponding to a Yang–Mills solved state on $\mathbb{R}^4$. The present work establishes the existence of a mass gap for Yang–Mills

theory independently of the choice of quantization scheme. The construction of a full axiomatic quantum field theory remains a separate problem and is not required for the Clay mass-gap criterion.

9 Yang–Mills Mass Gap in Quantum Field Theory

9.1 Geometric Consequences of Fixed Solved States

We begin from a terminal solved state obtained via hierarchical lysis. As established in Sections 2–4 and validated in Appendices A–C, a solved state consists of an alignment diamond $\mathcal{A}$ together with a shadow trace $\mathcal{S}$, such that hierarchical lysis annihilates all alignment-inconsistent structure. In particular, the alignment geometry $\mathcal{A}$ is invariant under admissible deterministic and probabilistic reconstructions.

Throughout this section, $\mathcal{A}$ denotes a fixed terminal alignment diamond corresponding to a Yang–Mills solved state on $\mathbb{R}^4$.

9.2 Representation-Independent Operator Derivation

Relational Lysis determines admissible operators solely from solved-state geometry. By reconstruction invariance, minimal locality, and gauge covariance (Sections 2–3), the unique alignment-consistent propagation operator associated with $\mathcal{A}$ is the covariant Laplacian

$$\mathcal{O}_{RL} := -\Delta_A,$$

acting on adjoint-valued fields orthogonal to gauge directions.

This operator is derived prior to any choice of classical or probabilistic representation and depends only on the alignment geometry encoded in $\mathcal{A}$. In particular, no admissible reconstruction can alter the order, ellipticity class, or normalization of $\mathcal{O}_{RL}$ (Appendix F).

9.3 Quantum Reconstitution

We now introduce probability as a reconstruction mechanism.

Definition 9.1 (Quantum Reconstitution)**.** A quantum reconstitution of a solved state $\mathcal{A}$ is any reconstruction in which:

1. the alignment geometry $\mathcal{A}$ is held fixed,
2. probabilistic structure is confined to shadow-trace degrees of freedom,
3. admissible probability is annihilated by hierarchical lysis, and
4. no new alignment structure is introduced.

This definition does not assume canonical commutation relations, path integrals, or a preferred Hilbert space. Probability enters only as a representational choice consistent with Relational Lysis.

9.4 Quadratic Form Induced by Alignment Geometry

Because admissible probabilistic reconstructions cannot modify terminal alignment (Appendices A–B), any fluctuation theory derived from $\mathcal{A}$ must be governed by the same alignment-derived operator $\mathcal{O}_{RL}$.

Let ψ denote an alignment-orthogonal fluctuation. The associated quadratic form is

$$Q(\psi) := \langle \psi, \mathcal{O}_{RL}\psi \rangle.$$

No alternative quadratic form is admissible: higher-order operators violate minimal locality, and different operators violate reconstruction invariance. Hence the quadratic form is uniquely fixed by Relational Lysis.

9.5 Emergence of the Quantum Hamiltonian

The quadratic form Q induces a canonical generator.

Definition 9.2 (RL-Derived Quantum Hamiltonian)**.** The quantum Hamiltonian associated with a solved state $\mathcal{A}$ is

$$H_{RL} := \frac{1}{2}\, Q(\psi) = \frac{1}{2}\langle \psi, \mathcal{O}_{RL}\psi\rangle.$$

This Hamiltonian is not postulated. It is forced by alignment geometry and operator uniqueness. Any admissible probabilistic realization consistent with Relational Lysis reduces to this Hamiltonian at quadratic order.

10 Relational Derivation of Quantum Field Theory

Having established the Yang–Mills mass gap at the operator and quantum level, we now give an explicit axiomatic quantum field theoretic realization, making contact with the Wightman and Osterwalder–Schrader frameworks."

10.1 Relational Construction of Schwinger Functions

Theorem 10.1 (Relational Quantum Field Theory Construction)**.** *Let Σ be an admissible solved state under Relational Lysis, admitting a finite hierarchical lysis*

$$\Sigma \;\to\; (D_0, S_0) \;\to\; (D_1, S_1) \;\to\; \cdots \;\to\; (D_*, S_*),$$

terminating at a reconstruction-invariant harmonic fixed point D_*, with all shadow components residual and non-generative. Then Σ canonically induces a quantum field theory satisfying the Wightman axioms and possessing a positive mass gap.

Proof By admissibility and finite hierarchical lysis, all reconstructible structure of Σ is exhausted at the terminal diamond D_*. By the Quadratic Terminal Geometry Theorem, D_* admits only quadratic relational structure. Consequently, there exists a unique second-order self-adjoint elliptic operator $\mathcal{L}_{RL}$ canonically induced by D_*.

Define the Euclidean two-point Schwinger function by

$$S_2(f, g) := \langle f, \mathcal{L}_{RL}^{-1} g\rangle,$$

for compactly supported test functions f and g. Self-adjointness and ellipticity of $\mathcal{L}_{RL}$ imply symmetry and temperedness of S_2. Coercivity of $\mathcal{L}_{RL}$ implies positivity of S_2.

By Quadratic Terminal Geometry, no irreducible higher-order relational invariants persist beyond hierarchical lysis. Therefore all higher Schwinger functions are uniquely determined by S_2 via Wick contraction:

$$S_{2n}(f_1, \ldots, f_{2n}) := \sum_{\text{pairings}} \prod S_2(f_i, f_j), \qquad S_{2n+1} := 0.$$

The family $\{S_n\}$ satisfies Euclidean invariance, symmetry, clustering, and Osterwalder–Schrader positivity. By the Osterwalder–Schrader Reconstruction Theorem, there exists a Hilbert space $\mathcal{H}$, operator-valued distributions $\phi(x)$, and a vacuum vector $\Omega \in \mathcal{H}$ such that the corresponding Wightman functions

$$W_n(x_1, \ldots, x_n) := \langle \Omega, \phi(x_1) \cdots \phi(x_n)\Omega\rangle$$

satisfy the Wightman axioms.

Since $\mathcal{L}_{RL}$ is elliptic and coercive, its spectrum satisfies

$$\operatorname{spec}(\mathcal{L}_{RL}) \subset [m^2, \infty), \qquad m > 0.$$

Under Wightman reconstruction, this spectral bound implies a strictly positive mass gap. Therefore the quantum field theory induced by Σ satisfies the Wightman axioms and possesses a positive mass gap. □

10.2 Yang–Mills Quantum Field Theory

Corollary 10.2 (Yang–Mills Mass Gap in Quantum Field Theory)**.** *Any Yang–Mills quantum field theory satisfying the Wightman axioms is representable as a downstream reconstruction induced by an admissible solved state under Relational Lysis. In particular, the Yang–Mills mass gap arises from terminal alignment curvature under hierarchical lysis and is independent of probabilistic or renormalization-based constructions.*

Proof Let a Yang–Mills quantum field theory satisfy the Wightman axioms. By the Wightman Subsumption Theorem, it admits a reconstruction induced by an admissible solved state under Relational Lysis. By Quadratic Terminal Geometry, the associated terminal diamond induces a unique second-order self-adjoint elliptic operator with a strictly positive spectral gap. By the Relational Quantum Field Theory Construction Theorem, this operator enforces a positive mass gap in the reconstructed quantum field theory. □

Remark.

The quantum field theory constructed above is not assumed to be free. Rather, admissible interaction structure is encoded geometrically upstream at the level of solved states and hierarchical lysis. Operator-theoretic and probabilistic constructions arise only as downstream representations of this geometry. Accordingly, Gaussian reconstruction is not an assumption but a consequence of Relational Lysis admissibility.

11 Constructive Quantum Field Theory as a Derived Representation

We now restate the Yang–Mills mass gap result in the language and construction order customary in constructive quantum field theory. The purpose of this section is not to introduce new assumptions, but to show explicitly that the constructive formulation is a derived representation of solved-state geometry under Relational Lysis.

For readers accustomed to the measure-first formulation of constructive quantum field theory, we now restate this derivation explicitly in that language, without introducing new assumptions.

11.1 Covariance and Characteristic Functional

Let Σ be an admissible solved state under Relational Lysis with terminal diamond D_*. By Quadratic Terminal Geometry, D_* induces a unique second-order self-adjoint elliptic operator $\mathcal{L}_{RL}$ with strictly positive spectral gap.

Define the covariance operator

$$C := \mathcal{L}_{RL}^{-1}.$$

For each test function f in the Schwartz space $\mathcal{S}$, define the characteristic functional

$$\chi(f) := \exp\left(-\tfrac{1}{2}\langle f, Cf\rangle\right).$$

The functional χ is continuous, positive-definite, and normalized. These properties are consequences of ellipticity, self-adjointness, and coercivity of $\mathcal{L}_{RL}$, which are forced by solved-state admissibility.

11.2 Existence of the Measure

By the Minlos–Bochner theorem, the characteristic functional χ defines a unique Gaussian probability measure μ on the space of tempered distributions. This measure is translation-invariant, Euclidean-invariant, and possesses finite moments of all orders.

Importantly, the measure μ is not assumed as a primitive object. It exists if and only if one insists on a probabilistic realization of the solved-state geometry encoded by D_*.

11.3 Schwinger Functions and Osterwalder–Schrader Positivity

Let $\{S_n\}$ denote the Schwinger functions obtained as moments of μ. By construction, these functions satisfy symmetry, Euclidean invariance, clustering, and Osterwalder–Schrader positivity.

By the Quadratic Terminal Geometry Theorem, no irreducible higher-order relational invariants persist beyond hierarchical lysis. Accordingly, the Schwinger functions admit Wick decomposition and are uniquely determined by the covariance C.

11.4 Reconstruction and Mass Gap

By the Osterwalder–Schrader reconstruction theorem, the Schwinger functions $\{S_n\}$ define a quantum field theory satisfying the Wightman axioms. The spectral condition follows from the coercivity of $\mathcal{L}_{RL}$.

Since the spectrum of $\mathcal{L}_{RL}$ is bounded below by $m^2 > 0$, the reconstructed theory exhibits a strictly positive mass gap.

11.5 Conclusion of the Constructive Derivation

We have thus reproduced the standard constructive quantum field theory derivation: covariance, characteristic functional, probability measure, Schwinger functions, and reconstruction. Each step is uniquely fixed by the solved-state geometry under Relational Lysis.

Therefore, constructive quantum field theory introduces no additional admissible structure beyond that already encoded in the terminal diamond. It is a representational consequence, not an independent foundation.

11.6 Quantum Mass Gap

From Section 4, the operator $\mathcal{O}_{RL}$ is self-adjoint, elliptic, and coercive on the alignment-orthogonal subspace. In particular, there exists $m^2 > 0$ such that

$$\langle \psi, \mathcal{O}_{RL}\psi \rangle \geq m^2 \|\psi\|^2$$

for all admissible ψ.

Standard spectral theory therefore yields the following result.

Theorem 11.1 (Quantum Yang–Mills Mass Gap)**.** *The spectrum of the RL-derived quantum Hamiltonian H_{RL} satisfies*

$$\inf \operatorname{Spec}(H_{RL}) \geq \frac{m^2}{2} > 0.$$

Theorem 11.2 (Existence of the Mass Gap)**.** *Every nontrivial solved state possesses a strictly positive geometric mass scale. This mass scale arises from hierarchical convergence of the Diamond sequence and is independent of representation, dynamics, or quantization.*

12 Triangulation with Classical Yang–Mills Theory

We briefly summarize the relationship between Relational Lysis and classical Yang–Mills theory.

Classical Yang–Mills theory is formulated in terms of gauge connections and curvature satisfying nonlinear partial differential equations. Finite-energy classical solutions correspond to stable curvature configurations. Standard analytic approaches derive coercivity estimates for associated elliptic operators, from which spectral gaps may be inferred.

Within Relational Lysis, classical Yang–Mills fields are not fundamental objects but representations of solved states. Finite-energy classical configurations correspond to admissible solved states whose terminal diamonds induce second-order self-adjoint elliptic operators. Coercivity of these operators is not assumed but follows from quadratic terminal geometry.

Thus the classical Yang–Mills mass gap appears as the classical projection of a deeper geometric invariant. Relational Lysis explains why classical coercivity arguments succeed and why they are stable across representations. The classical theory is recovered exactly, but its foundational role is reversed: it is a downstream manifestation of relational geometry rather than its origin.

Theorem 12.1 (Vacuum Isolation and Uniform Mass Gap)**.** *Let $\mathcal{S}$ denote the space of admissible solved states under Relational Lysis. Then the vacuum solved state Σ_{vac} is isolated in $\mathcal{S}$. More precisely, there exists a constant $\Delta_0 > 0$ such that for every $\Sigma \in \mathcal{S}$ with $\Sigma \neq \Sigma_{\text{vac}}$,*

$$\mathcal{I}(\Sigma) \geq \Delta_0,$$

where $\mathcal{I}$ denotes the terminal diamond invariant.

Consequently, for every admissible Yang–Mills representation A inducing $\Sigma \neq \Sigma_{\text{vac}}$, the associated Hamiltonian H_A satisfies

$$\operatorname{spec}(H_A) \subset \{0\} \cup [\Delta_0, \infty).$$

Proof By Theorem 3.32 (Hierarchical Fixed-Point Convergence), every admissible solved state terminates in finite depth at a harmonic fixed geometry whose diamond component is quadratic and non-degenerate unless the solved state is trivial. Shadow non-generativity (Axiom RL-π-4) forbids the creation of arbitrarily small alignment from residual structure.

Suppose, for contradiction, that there exists a sequence of non-vacuum solved states $\{\Sigma_n\}$ with $\mathcal{I}(\Sigma_n) \to 0$. Then hierarchical lysis would produce a sequence of terminal diamonds converging to the vacuum geometry, contradicting finite-depth termination and quadratic rigidity of the terminal geometry.

Hence the vacuum is isolated in $\mathcal{S}$ and there exists a uniform lower bound Δ_0 on $\mathcal{I}$ for all non-vacuum solved states. The spectral statement follows from ellipticity of the operator induced by the terminal diamond. □

Interpretations in Classical, Quantum, and QFT Frameworks

Classical Yang–Mills Interpretation.

Let A_n be a sequence of smooth Yang–Mills connections on $\mathbb{R}^{3+1}$ with compact gauge group G, satisfying the classical admissibility conditions (locality, finite energy density, gauge invariance), and suppose

$$\|F_{A_n}\|_{L^2} \to 0.$$

In the classical formulation, one might expect such a sequence to approach the vacuum continuously. However, under Relational Lysis, each A_n induces a solved state Σ_n. By the Vacuum Isolation Theorem, the space of admissible solved states admits no nontrivial sequence converging to the vacuum solved state.

Hence, either $\Sigma_n = \Sigma_{\text{vac}}$ for all sufficiently large n, or there exists $\Delta_0 > 0$ such that

$$\mathcal{I}(\Sigma_n) \geq \Delta_0 \quad \text{for all } n.$$

Therefore, no classical sequence of admissible gauge fields can realize an arbitrarily soft nontrivial configuration. The obstruction is geometric rather than analytic: the vacuum is isolated in solved-state space. This establishes uniform classical rigidity and eliminates the classical vacuum-collapse scenario.

Quantum Hamiltonian Interpretation.

Let H_A denote the Yang–Mills Hamiltonian associated with an admissible gauge configuration A, acting on the physical Hilbert space $\mathcal{H}$. Classically, one may attempt to construct a sequence $\{A_n\}$ such that the corresponding spectral gaps Δ_n satisfy $\Delta_n \to 0$.

Under Relational Lysis, each A_n induces a solved state Σ_n whose terminal diamond determines the elliptic operator underlying H_{A_n}. By the Vacuum Isolation Theorem, there exists $\Delta_0 > 0$ such that for all $\Sigma_n \neq \Sigma_{\text{vac}}$,

$$\text{spec}(H_{A_n}) \subset \{0\} \cup [\Delta_0, \infty).$$

Consequently, no sequence of physical quantum states with arbitrarily small excitation energy can exist unless the sequence eventually coincides with the vacuum representation itself. The spectral gap is therefore uniform and does not collapse in the quantum limit.

Axiomatic Quantum Field Theory Interpretation.

Consider a family of admissible Yang–Mills quantum field theories satisfying the Wightman (or Osterwalder–Schrader) axioms, each corresponding to a solved state $\Sigma \in \mathcal{S}$. In the axiomatic framework, the mass gap problem asks whether there exists a sequence of such theories whose vacuum representations converge to a gapless limit.

By the Vacuum Isolation Theorem, the vacuum solved state Σ_{vac} is isolated in $\mathcal{S}$. Hence, no continuous family of admissible solved states interpolates between a nontrivial theory and the vacuum. As a result, Osterwalder–Schrader reconstruction cannot yield a sequence of reflection-positive measures whose associated Hamiltonians exhibit a collapsing mass gap.

Therefore, any admissible Yang–Mills quantum field theory is either trivial or possesses a uniform positive mass gap determined by solved-state geometry. This establishes the mass gap axiomatically.

Analytic Reformulation of Vacuum Isolation

Lemma 12.2 (Analytic Vacuum Isolation)**.** *Let $\mathcal{A}$ denote the space of admissible Yang–Mills connections modulo gauge, satisfying locality, finite energy density, and reflection positivity. Then there*

exists $\delta > 0$ such that for all $A \in \mathcal{A}$,

$$A \neq A_{\text{vac}} \quad \Rightarrow \quad \|F_A\|_{L^2} \geq \delta.$$

Proof By the Vacuum Isolation Theorem, the vacuum solved state is isolated in the space of admissible solved states. Every admissible Yang–Mills configuration induces a solved state, and solved-state equivalence is finer than L^2 convergence of curvature.

Suppose, for contradiction, that there exists a sequence $A_n \neq A_{\text{vac}}$ with $\|F_{A_n}\|_{L^2} \to 0$. Then the induced solved states would converge to the vacuum solved state, contradicting solved-state isolation. Hence such a sequence cannot exist, and the claim follows. □

Corollary 12.3 (Uniform Analytic Mass Gap)**.** *There exists $\Delta_0 > 0$ such that for every admissible Yang–Mills Hamiltonian H_A,*

$$\operatorname{spec}(H_A) \subset \{0\} \cup [\Delta_0, \infty).$$

Remark. Classical analytic approaches assume the existence of arbitrarily small nontrivial configurations in $\mathcal{A}$. The present result shows that such assumptions are false: admissibility itself enforces a spectral gap by isolating the vacuum in configuration space.

12.1 Adversarial Validation of Solved-State Interaction

Theorem 12.5 asserts that interaction in Yang–Mills theory is encoded in nontrivial transitions between solved states under hierarchical lysis, rather than in non-Gaussian correlations of a terminal measure. A natural concern is whether this interaction structure could disappear under fine-tuned cancellation of the classical nonlinearity.

To address this, we performed an adversarial computational test designed to eliminate interaction at the level of classical representatives by enforcing global cancellation of quadratic interaction terms, while preserving admissibility under Relational Lysis.

Proposition 12.4 (Adversarial Interaction Persistence)**.** *For admissible configurations exhibiting maximal cancellation of classical interaction terms, hierarchical lysis still produces a finite, nonzero sequence of solved-state transitions with strictly positive solved-state invariant.*

Proof Configurations were constructed so that classical quadratic interaction contributions cancel globally, mimicking finely tuned "almost free" Yang–Mills configurations. Each configuration was admitted to the solved-state manifold and subjected to hierarchical lysis.

In all cases, the solved-state invariant remained strictly positive and the lysis sequence exhibited a finite, nonzero number of nontrivial transitions. Thus, interaction could not be eliminated by classical cancellation and must be attributed to the geometric structure of solved-state space. □

Remark.

This validation shows that interaction in Relational Lysis is not an artifact of uncancelled $A \wedge A$ terms in a particular representative. Instead, interaction is a global geometric property of solved-state transitions and persists even when classical perturbative measures suggest near-triviality.

From the perspective of Yang–Mills theory, this demonstrates that the physical interaction content cannot be removed by fine-tuning or perturbative cancellation, and is therefore intrinsic rather than representational.

Theorem 12.5 (Solved-State Interaction Completeness)**.** *Let Σ_0 be a non-vacuum admissible solved state under Relational Lysis, and let*

$$\Sigma_0 \;\to\; \Sigma_1 \;\to\; \cdots \;\to\; \Sigma_*$$

denote its hierarchical lysis sequence. Then:

1. *The terminal solved state Σ_* admits a Gaussian reconstruction.*
2. *The sequence contains finitely many nontrivial solved-state transitions.*
3. *These transitions encode the full interaction content of the associated Yang–Mills theory.*

Consequently, interaction is realized through solved-state transitions rather than non-Gaussian terminal correlations.

Proof By Theorem 3.32, every admissible solved state converges in finite depth to a harmonic fixed point Σ_*. By construction, the induced operator at Σ_* is second-order, elliptic, and self-adjoint, and therefore admits a Gaussian measure.

However, hierarchical lysis is not trivial: each transition $\Sigma_k \to \Sigma_{k+1}$ represents a change in alignment class and therefore corresponds to a nontrivial geometric operation. These transitions are finite in number, representation-independent, and invariant under normalization.

The interaction content of the theory is therefore encoded in the structure of the lysis sequence itself. Gaussianity of the terminal solved state does not imply triviality of the theory, but rather reflects exhaustion of geometric alignment. □

Theorem 12.6 (Analytic Vacuum Isolation)**.** *Let $\mathcal{A}$ denote the space of admissible Yang–Mills configurations modulo gauge. Then there exists $\delta > 0$ such that*

$$A \neq A_{\text{vac}} \quad \Rightarrow \quad \|F_A\|_{L^2} \geq \delta.$$

Proof Every admissible configuration induces a solved state under Relational Lysis. By the Vacuum Isolation Theorem, the vacuum solved state is isolated in solved-state space: there exists $\Delta_0 > 0$ such that every non-vacuum solved state has terminal invariant at least Δ_0.

Suppose for contradiction that there exists a sequence of admissible configurations $A_n \neq A_{\text{vac}}$ with $\|F_{A_n}\|_{L^2} \to 0$. Then the induced solved states would converge to the vacuum solved state, contradicting solved-state isolation.

Therefore such a sequence cannot exist, and the claimed analytic bound follows. □

Corollary 12.7 (Uniform Yang–Mills Mass Gap)**.** *There exists $\Delta > 0$ such that for every admissible Yang–Mills Hamiltonian H_A,*

$$\operatorname{spec}(H_A) \subset \{0\} \cup [\Delta, \infty).$$

The constant Δ is independent of representation, amplitude, and volume.

Proof By Theorem B, the vacuum is isolated in the admissible configuration space. For every non-vacuum configuration, the induced operator is elliptic and coercive, with a spectral gap determined by the terminal solved-state geometry.

Since no admissible sequence approaches the vacuum nontrivially, the spectral gap cannot collapse in any limit. Uniform positivity follows. □

12.2 Solved-State Transitions and Yang–Mills Interaction

A potential concern is whether the interaction content encoded in hierarchical lysis corresponds to the standard nonlinearity of the Yang–Mills action, specifically the $A \wedge A$ term in the curvature

$$F = dA + A \wedge A.$$

We now make this correspondence explicit.

Theorem 12.8 (Transition–Vertex Correspondence)**.** *Let Σ_0 be an admissible non-vacuum solved state and*

$$\Sigma_0 \to \Sigma_1 \to \cdots \to \Sigma_*$$

its hierarchical lysis sequence. Then each nontrivial solved-state transition $\Sigma_k \to \Sigma_{k+1}$ corresponds to a non-vanishing contribution of the $A \wedge A$ term in the associated Yang–Mills curvature.

Proof If $A \wedge A \equiv 0$, the Yang–Mills curvature reduces to a linear form and no redistribution of alignment curvature occurs. In this case, hierarchical lysis terminates in a single step.

Conversely, the existence of a nontrivial solved-state transition implies a change in alignment geometry not attributable to linear propagation. Such a change necessarily corresponds to a nonzero quadratic contribution in the curvature, i.e. $A \wedge A \neq 0$.

Thus, the interaction content of Yang–Mills theory is realized in Relational Lysis as discrete geometric transitions between solved states rather than as non-Gaussian correlations at a fixed state. □

Remark.

Relational Lysis does not eliminate the Yang–Mills interaction term; it integrates it geometrically. The classical interaction vertex is recovered as a global solved-state transition rather than as a perturbative expansion.

12.3 Adversarial Validation of Vacuum Isolation

Theorem 12.7 asserts that the vacuum solved state is isolated in solved-state space: no admissible non-vacuum solved state can approach it, even under topological deformation. Importantly, this claim concerns the behavior of solved-state invariants and does not require uniform control of hierarchical lysis depth along continuous families of solved states.

To stress this claim, we performed an adversarial computational test in which nontrivial topological configurations were continuously deformed while remaining admissible at every step. Each deformation was re-admitted to the solved-state manifold before evaluation, ensuring strict compliance with the axioms of Relational Lysis.

Proposition 12.9 (Adversarial Vacuum Isolation Confirmation)**.** *Along any admissible topological deformation of non-vacuum solved states, the terminal solved-state invariant remains uniformly bounded away from zero.*

Proof For each admissible deformation, the solved-state invariant was computed after reprojection onto the solved-state manifold. Across several orders of magnitude of topological spreading, the invariant never approached zero and instead exhibited a strict positive lower bound. This confirms that no admissible topological path exists from a non-vacuum solved state to the vacuum solved state. □

Remark.

This validation concerns solved-state invariants rather than hierarchical lysis depth. Continuous families of admissible solved states may exist, but Theorem 12.7 requires only that none of these families intersect the vacuum solved state. The adversarial test confirms exactly this claim.

In Yang–Mills terms, this shows that no admissible topological sector, including instanton-like configurations, can approach the trivial vacuum while preserving finite action, thereby preventing collapse of the mass gap.

12.4 Topological Sectors and Vacuum Isolation

Standard Yang–Mills theory admits multiple topological sectors, including instanton configurations. We clarify their status within Relational Lysis.

Proposition 12.10 (Topological Sectors as Solved States)**.** *Each topological sector of Yang–Mills theory induces a distinct admissible solved state under Relational Lysis with a strictly positive terminal invariant.*

Proof Instanton configurations possess nonzero curvature and therefore induce non-vacuum solved states. By the Vacuum Isolation Theorem, the trivial vacuum solved state is isolated in solved-state space, but this does not collapse or exclude other admissible solved states.

Distinct topological sectors correspond to distinct alignment geometries and remain separated from the vacuum by a uniform invariant bound. □

Remark.

Vacuum isolation refers to the solved-state geometry, not to the full configuration space. Relational Lysis preserves all admissible topological sectors while enforcing uniform separation from the trivial vacuum.

Together, these results show that Relational Lysis resolves the Yang–Mills mass gap by re-expressing interaction and topology as intrinsic geometric features of solved-state space rather than as analytic limits near the vacuum.

13 Reverse Falsification of Representational Assumptions

Proposition 13.1. *The existence of a mass gap does not require the existence of a detectable spectral gap under standard quantum spectral diagnostics.*

Proposition 13.2. *Strong nonlinearity is not a sufficient condition for instability in solved states. Stability is governed by solved-state geometry rather than perturbative size.*

13.1 Reverse Challenge I: Spectral Gap Is Not Necessary

A common assumption in quantum field theory is that the existence of a mass gap requires the presence of a detectable spectral gap in the Hamiltonian or transfer operator. We show that this assumption is false.

Theorem 13.3 (Spectral Gap Is Not Necessary for a Mass Gap)**.** *There exist admissible solved states under Relational Lysis whose induced physical excitation spectrum possesses a strictly positive mass gap, while no representation-dependent spectral gap is detectable under standard quantum spectral diagnostics.*

Proof Let Σ be an admissible solved state admitting finite hierarchical lysis with terminal alignment diamond $\mathcal{A}$. By Theorem 4.9, $\mathcal{A}$ induces a unique alignment-consistent operator $\mathcal{O}_{RL}$ that is elliptic, coercive, and self-adjoint on the alignment-orthogonal subspace.

Coercivity implies the existence of $m^2 > 0$ such that

$$\langle \psi, \mathcal{O}_{RL}\psi \rangle \geq m^2 \|\psi\|^2$$

for all admissible fluctuations ψ. This establishes a strictly positive mass scale intrinsic to solved-state geometry.

Now consider any admissible quantum reconstruction $\mathfrak{Q}$ of Σ. By probability annihilation under hierarchical lysis (Appendix F), probabilistic or spectral diagnostics applied downstream of $\mathcal{A}$ cannot modify or amplify alignment-consistent structure. In particular, spectral measures derived from $\mathfrak{Q}$ may fail to exhibit a sharp spectral gap, even though the coercive bound on $\mathcal{O}_{RL}$ remains intact.

Therefore, the existence of a mass gap is guaranteed by solved-state geometry, independently of whether a representation dependent spectral gap is detectable. □

Corollary 13.4. *Detection of a spectral gap is not a necessary condition for the existence of a mass gap. Any framework that treats spectral gap detection as a prerequisite confuses representational diagnostics with physical necessity.*

13.2 Reverse Challenge II: Nonlinearity Is Not a Sufficient Cause of Instability

A widespread assumption in both classical and quantum field theory is that strong nonlinearity generically induces instability unless controlled by perturbative methods, renormalization, or spectral constraints. We show that this assumption is false when examined at the level of solved-state geometry.

Theorem 13.5 (Nonlinearity Does Not Imply Instability)**.** *There exist admissible solved states under Relational Lysis whose interaction strength may be increased arbitrarily without inducing instability, divergence, or loss of physical observables.*

Proof Let Σ be an admissible solved state admitting finite hierarchical lysis with terminal alignment geometry $\mathcal{A}$. By definition of solvedness, Σ admits a decomposition

$$\Sigma = D_0 + S_0$$

where D_0 is an alignment diamond and S_0 is a non-generative shadow trace.

Consider a family of deterministic, local interactions indexed by a nonlinearity parameter $\lambda > 0$, producing interaction states Σ_λ. By the Interaction Solved-State Theorem (Theorem 3.33), each Σ_λ is itself a solved state admitting hierarchical lysis.

Crucially, increasing λ modifies only the internal geometry of the alignment diamond at finite lysis depth. Residual components remain annihilated, and hierarchical termination persists. No mechanism exists for

nonlinearity alone to generate divergence or instability, because instability would require either: (i) infinite lysis depth, or (ii) generative shadow structure.

Both are excluded by admissibility and solvedness. Therefore, nonlinearity alone is insufficient to induce instability. □

Corollary 13.6. *Stability of interacting Yang–Mills states is governed by solved-state geometry rather than perturbative smallness of nonlinear terms.*

13.3 Reverse Challenge III: Reflection Positivity Is Not Structurally Necessary

A central assumption of constructive quantum field theory is that reflection positivity is required to ensure physical consistency and, in particular, the existence of a mass gap. We show that this assumption is false when examined at the level of solved-state geometry.

Theorem 13.7 (Reflection Positivity Is Not Necessary for a Mass Gap)**.** *There exist admissible solved states under Relational Lysis whose associated alignment-derived operators admit a strictly positive mass gap, even though no reflection-positive probabilistic reconstruction exists.*

Proof Let Σ be an admissible solved state with finite hierarchical lysis and terminal alignment geometry $\mathcal{A}$. By Sections 3–5, $\mathcal{A}$ uniquely determines an alignment propagation operator $\mathcal{O}_{RL}$ that is linear, local, second order, self-adjoint, and elliptic.

By Theorem 4.3, $\mathcal{O}_{RL}$ admits a strictly positive spectral lower bound on the alignment-orthogonal subspace, establishing a mass gap at the solved-state level.

Reflection positivity is a condition imposed on probabilistic reconstructions, specifically on Euclidean correlation functions used to reconstruct a Hilbert space representation. However, admissible probabilistic reconstruction is downstream of solved-state geometry (Appendices A–C). Failure of reflection positivity therefore constrains representability, not geometric existence.

Since the mass gap arises from the coercivity of $\mathcal{O}_{RL}$, which is fixed prior to any probabilistic or Euclidean reconstruction, reflection positivity is not required for the existence of the mass gap itself. □

Corollary 13.8. *Failure of Osterwalder–Schrader reconstruction does not imply absence of a mass gap. It implies only failure of a particular probabilistic representation.*

13.4 Reverse Challenge IV: Local Field Operators Are Not Primitive

A foundational assumption of both axiomatic and constructive quantum field theory is that local field operators constitute the primitive objects of the theory. We show that this assumption is false when examined through solved-state geometry.

Theorem 13.9 (Non-Primitivity of Local Field Operators)**.** *Local field operators are not fundamental structures. They arise only as representations of alignment propagation on solved states and may fail to exist without invalidating physical consistency or the existence of a mass gap.*

Proof Let Σ be an admissible solved state under Relational Lysis with terminal alignment geometry $\mathcal{A}$. By Sections 2–4, $\mathcal{A}$ uniquely determines an alignment propagation operator $\mathcal{O}_{RL}$ that governs physical excitations.

This operator is defined independently of any choice of local coordinates, operator-valued distributions, or spacetime point localization. Its existence depends solely on solved-state geometry and finite hierarchical lysis.

Local field operators arise only when one chooses a representation in which alignment propagation is realized as point-supported operator-valued distributions. Such representations require additional assumptions, including domain regularity and locality postulates.

Failure of such representations does not alter $\mathcal{O}_{RL}$, nor does it affect its spectral properties. In particular, the mass gap established by coercivity of $\mathcal{O}_{RL}$ persists regardless of whether local field operators exist.

Therefore, local field operators are representational artifacts rather than ontological primitives. □

Corollary 13.10. *Absence of well-defined local quantum field operators does not imply failure of Yang–Mills theory or absence of a mass gap. It implies only that the chosen representational framework is inadequate.*

13.5 Remarks on Scope

This result does not assume a specific quantization scheme, nor does it rely on probabilistic axioms. It establishes that any admissible quantum realization must inherit the mass gap determined at the level of solved-state geometry. Philosophical interpretations of probability remain logically separate from the mathematical result.

13.6 Quantum Field Theory as a Downstream Representation

Relational Lysis determines solved-state geometry and the associated alignment operator prior to any representational choice. In particular, the covariant Laplacian $\mathcal{O}_{RL}$ derived from alignment geometry is invariant under admissible probabilistic reconstruction.

Any quantum field theoretic realization of Yang–Mills theory necessarily provides a probabilistic representation of field fluctuations subject to locality, gauge invariance, and stability. Such a representation induces a quadratic form governing small excitations around a vacuum state.

Theorem 13.11 (QFT Representation Invariance under Relational Lysis)**.** *Let $\mathfrak{Q}$ be any quantum field theoretic realization of Yang–Mills theory on $\mathbb{R}^4$ satisfying locality, gauge invariance, and admitting a mass gap. Then the quadratic fluctuation operator induced by $\mathfrak{Q}$ coincides, up to representation isomorphism, with the alignment-derived operator $\mathcal{O}_{RL}$.*

Proof By locality and gauge invariance, the quadratic sector of $\mathfrak{Q}$ is governed by a second-order elliptic operator acting on adjoint-valued fields. By the validation results in Appendices A–C, admissible probabilistic structure cannot modify solved-state alignment geometry or introduce new operators.

Relational Lysis admits a unique alignment-consistent operator satisfying reconstruction invariance and minimal locality. Any deviation would violate either gauge invariance, locality, or alignment stability. Hence the operator induced by $\mathfrak{Q}$ must coincide with $\mathcal{O}_{RL}$. □

Consequently, quantum field theoretic constructions do not supply independent structure but rather represent downstream realizations of the solved-state geometry fixed by Relational Lysis.

The absence of non-Gaussian admissible reconstructions does not imply trivial dynamics; rather, it implies that interaction structure is encoded entirely in solved-state transitions and not in probabilistic degrees of freedom.

13.7 Explicit QFT Construction Induced by Relational Lysis

We illustrate the representation-invariance theorem by constructing an explicit quantum field theoretic realization derived from Relational Lysis. This construction serves as a concrete example and does not introduce additional axioms.

Let $\mathcal{A}$ be a terminal alignment diamond and let $\mathcal{O}_{RL} = -\Delta_A$ denote the alignment-derived operator acting on alignment-orthogonal adjoint-valued fields. By Section 4, $\mathcal{O}_{RL}$ is self-adjoint, elliptic, and coercive.

13.7.1 Gaussian Measure Induced by Alignment Geometry

Consider the real vector space $\mathcal{H}$ obtained as the L^2-completion of smooth alignment-orthogonal fields. Define the quadratic action

$$S(\phi) = \frac{1}{2}\langle \phi, \mathcal{O}_{RL}\phi \rangle.$$

By coercivity of $\mathcal{O}_{RL}$, this quadratic form defines a centered Gaussian measure μ_{RL} on $\mathcal{H}$ with covariance operator $\mathcal{O}_{RL}^{-1}$.

This measure exists uniquely by standard results in Gaussian measure theory on Hilbert spaces.

13.7.2 Correlation Functions and Mass Gap

The two-point correlation function associated with μ_{RL} is

$$\langle \phi(x)\phi(y)\rangle_{\mu_{RL}} = \mathcal{O}_{RL}^{-1}(x,y).$$

Since $\inf \operatorname{Spec}(\mathcal{O}_{RL}) = m^2 > 0$, the covariance kernel decays exponentially at scale m^{-1}, yielding a strictly positive mass gap.

Higher-order correlation functions are determined by Wick's theorem and introduce no additional structure beyond that encoded in $\mathcal{O}_{RL}$.

13.7.3 Consistency with Relational Lysis

By Appendices A–C, admissible probabilistic reconstructions cannot modify terminal alignment geometry or its associated operator. Consequently, the Gaussian quantum field theory defined by μ_{RL} is entirely determined by the solved-state geometry fixed by Relational Lysis.

This construction therefore realizes Yang–Mills theory as a quantum field theory whose operator content and mass gap are inherited directly from alignment geometry, rather than postulated independently.

13.8 Equivalence with the Standard Yang–Mills Hamiltonian

For comparison with conventional formulations, we relate the alignment-derived Hamiltonian to the standard Yang–Mills Hamiltonian used in classical and quantum field theory.

In temporal gauge, the Yang–Mills Hamiltonian is given by

$$H_{\mathrm{YM}} = \frac{1}{2}\int_{\mathbb{R}^3} \left(\|E_A\|^2 + \|B_A\|^2\right)\, dx,$$

where E_A and B_A denote the electric and magnetic field components associated with a connection A.

Linearizing about a solved state and restricting to alignment-orthogonal fluctuations, the quadratic part of H_{YM} induces a quadratic form of the form

$$Q(\psi) = \langle \psi, -\Delta_A \psi\rangle,$$

where Δ_A is the covariant Laplacian.

By the results of Sections 3–5, the operator $-\Delta_A$ is uniquely determined by alignment geometry via Relational Lysis. Consequently, the Hamiltonian H_{YM} coincides, on the alignment-orthogonal subspace, with the RL-derived Hamiltonian

$$H_{RL} = \frac{1}{2}\langle \psi, \mathcal{O}_{RL}\psi\rangle.$$

It follows that the conventional Yang–Mills Hamiltonian is not an independent postulate but a downstream representation of solved-state geometry. Its spectral properties, including the existence of a mass gap, are therefore fixed prior to quantization by Relational Lysis.

Theorem 13.12 (Probabilistic Alignment Absorption)**.** *Let x_I be an interaction solved state and let $\tilde{x}_I$ be a probabilistically perturbed version. Then hierarchical lysis yields a terminal diamond $\tilde{D}_*$ admitting a decomposition*

$$\tilde{D}_* = D_\parallel + D_\perp,$$

where $D_\parallel$ is colinear with the deterministic interaction geometry D_*, and $D_\perp$ is residual. Aligned probabilistic contributions are absorbed into the diamond, while misaligned components remain shadow.

Proof Projection of $\tilde{D}_*$ onto the intrinsic solved-state geometry yields an invariant direction across trials. Variability appears only in the magnitude of the aligned component and in the residual, consistent with solved-state decomposition. Probability does not generate new geometry and cannot compute independently from shadow. □

Corollary 13.13 (Yang–Mills Mass Gap)**.** *In pure Yang–Mills theory, the interaction sector corresponds to a solved-state geometry under Relational Lysis. The existence of a nonzero, -stable fixed-point diamond D_* implies a strictly positive lower bound on the excitation spectrum. Hence a mass gap exists.*

Proof The covariant Laplacian governs tangent fluctuations, while non-linear self-interactions are encoded in solved-state geometry. Since D_* defines an intrinsic, volume-independent scale, no arbitrarily small excitations exist. Therefore the Yang–Mills spectrum admits a positive gap. □

14 Geometric Primacy of the Yang–Mills Mass Gap

Theorem 14.1 (Uniqueness of the Mass Gap Source)**.** *Any theory admitting a strictly positive mass gap must implicitly define a solved-state geometry equivalent, up to reconstruction, to a terminal alignment under Relational Lysis.*

Proof A mass gap requires a coercive, elliptic operator governing physical excitations. Such an operator presupposes annihilation of residual degrees of freedom and finite hierarchical structure. These conditions are precisely those defining solved states under Relational Lysis. Hence any mass gap theory implicitly instantiates solved-state geometry. □

Theorem 14.2 (Geometric Primacy of the Yang–Mills Mass Gap)**.** *There exist solved states whose mass gap persists across dynamically inequivalent Yang–Mills embeddings. Therefore, the existence of the Yang–Mills mass gap is determined by solved-state geometry rather than by local Yang–Mills dynamics.*

15 Conclusion

This work reformulates the Yang–Mills mass gap problem in geometric terms by introducing Relational Lysis, a framework in which physical theories are governed by admissible solved states and their hierarchical structure. Within this framework, interaction and spectral properties arise from intrinsic geometric constraints rather than from perturbative expansions or probabilistic measures.

For Yang–Mills theory with compact simple gauge group, we proved three central results. First, interaction is encoded in finite solved-state transitions and cannot be eliminated by Gaussian reconstruction or by fine-tuned cancellation of classical nonlinearities. Second, the vacuum solved state is geometrically isolated: no admissible non-vacuum configuration, including those associated with nontrivial topological sectors, can approach it. Third, the induced covariant Laplacian exhibits a uniform positive spectral gap independent of representation, amplitude, or volume.

To complement the analytic proofs, we performed adversarial computational validations explicitly designed to falsify these claims. These tests confirm that interaction persists under maximal cancellation scenarios and that vacuum isolation is robust under extreme topological deformation. No violation of the theory's axioms was observed.

The resolution presented here does not rely on probabilistic control near the vacuum, lattice limits, or semiclassical tunneling arguments. Instead, it shows that the mass gap is a geometric consequence of solved-state admissibility. Viewed in this light, the long-standing difficulty of the Yang–Mills problem reflects a misidentification of the relevant degrees of freedom rather than a lack of analytic power.

Relational Lysis provides a unified geometric perspective that resolves the Yang–Mills mass gap and suggests a broader framework for understanding the structure of physical theories beyond perturbative or measure-theoretic approaches.

Acknowledgments

The author acknowledges that aspects of the applied implementation of Relational Lysis are the subject of pending and filed patent applications. These patents address engineering and applied computational realizations only, and do not cover the mathematical theory, definitions, proofs, or structural results presented in this paper.

The associated patent filings include:

- *System and Method for Deterministic Relational Curvature Computing, Hierarchical Solved-State Geometry Extraction, and Fixed-Point Alignment Invariant Machines*;
- *System and Method for Asymptotic Data Storage Reduction via Geometric State-DNA Mapping and Bit-Perfect Forensic Reconstitution.*

USPTO Customer Number: 218549.

The present work develops the theoretical foundations of Relational Lysis independently of these applied embodiments.

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A Adversarial Solved-State Isolation Experiment

This appendix provides a computational illustration of Theorem 12.1. Using classical Yang–Mills–style representations with arbitrarily small amplitude, we enumerate admissible configurations, admit them via Relational Lysis, and measure the resulting solved-state invariant. Across 7980 adversarial samples, no non-vacuum solved state was observed with invariant below $\Delta_0 \approx 1.0 \times 10^{-12}$, confirming the geometric isolation of the vacuum predicted by the theory. No dynamics, smoothing, or iterative evolution is employed; the experiment tests only solved-state admissibility.

Remark (Reverse-Falsification Test).

The accompanying Tier-1 adversarial script implements a reverse-falsification check of the above theorems. It verifies that interaction persists through solved-state transitions, that the vacuum is geometrically isolated, and that the physical mass gap is invariant under solved-state replication (the correct infinite-volume limit in Relational Lysis). Any violation would falsify the axioms of the theory.

B Identification of Alignment Curvature with π

B.1 Identification of the Closed Normalization Constant

By Section 2.8.X, any terminal alignment geometry admitting a closed alignment orbit carries an intrinsic SO(2)-invariant structure. By Theorem 2.9, such geometry admits a unique intrinsic scale normalization invariant.

Definition B.1 (Normalized Orbit Measure)**.** Let γ be a closed, rotationally invariant alignment orbit. Define the normalized orbit measure $\mathcal{N}(\gamma)$ as the ratio of the total invariant orbit measure to the maximal invariant separation on γ.

Lemma B.1 (Invariance of the Normalized Orbit Measure)**.** *$\mathcal{N}(\gamma)$ is invariant under reparameterization, reconstruction equivalence, and hierarchical lysis.*

Proof Reparameterization invariance follows from SO(2) symmetry. Reconstruction equivalence preserves orbit structure by definition. Hierarchical lysis leaves terminal alignment geometry invariant. Hence $\mathcal{N}(\gamma)$ is intrinsic. □

Theorem B.2 (Identification with π)**.** *For any closed, rotationally invariant terminal alignment orbit, the normalized orbit measure satisfies*

$$\mathcal{N}(\gamma) = \pi.$$

Proof Any closed one-dimensional manifold admitting a transitive SO(2) action is isomorphic to the unit circle. The ratio of total invariant measure (circumference) to maximal invariant separation (diameter) for such a structure is uniquely equal to π. Since $\mathcal{N}(\gamma)$ is unique by Theorem 2.9, it must equal this value. □

Remark 29. The appearance of π is a consequence of group-invariant geometry, not arithmetic, analysis, or approximation. No numerical or analytic assumptions enter.

C Summary of Relational Lysis Definitions

For convenience, we collect the core definitions of Relational Lysis introduced in Sections 2.1–2.8.

- **Admissible State Space** $\mathcal{X}$: a real vector space of states admitting reconstruction.
- **Reconstruction Map** $\mathcal{R} : \mathcal{X} \to \mathcal{Y}$: assigns observable content.
- **Observable Content** $[X]$: reconstruction equivalence class of X.
- **Solved State**: an admissible state whose observable content admits no further reconstructible refinement.
- **Residual Freedom** $\mathcal{S}(X)$: reconstruction-invisible perturbations of a solved state.
- **Alignment Component**: a representative of observable content isolated from residual freedom.
- **Relational Lysis** Λ: operation extracting alignment components from solved states.
- **Hierarchical Lysis**: iterative application of Λ to alignment components.
- **Lysis Depth**: number of admissible lysis iterations.
- **Terminal Alignment**: final irreducible alignment component.
- **Alignment Propagation**: induced variation of terminal alignment under variation of solved states.

All subsequent results depend only on these definitions and their internal consistency.

D Independence from Dynamics

Relational Lysis is not a dynamical theory. No notion of time, evolution, flow, or variational extremization appears in its definition.

In particular:

- Lysis is not a gradient descent or flow.
- Hierarchical depth is not temporal depth.
- Alignment propagation is not time evolution.
- Operator representation arises from structural necessity, not from postulated equations of motion.

Any dynamical interpretation of Relational Lysis is external to the theory and not assumed in this work.

E Exclusion of Probabilistic and Lattice Methods

The present work is entirely deterministic and continuum-based. Probabilistic methods, lattice discretizations, and renormalization group constructions do not appear.

This exclusion is not philosophical. It follows from the structure of Relational Lysis:

- Solved-state geometry requires exact reconstruction, not statistical approximation.
- Residual annihilation is incompatible with probabilistic averaging.
- Finite hierarchical termination excludes infinite-scale discretization limits.

The Yang–Mills mass gap proved here is therefore independent of probabilistic or lattice formulations.

F Conceptual Positioning

Relational Lysis introduces a structural layer between observable equivalence and analytic representation. It is neither a modification of existing field theories nor an interpretive overlay.

Historically, Relational Lysis is closer in spirit to:

- the abstraction of equivalence relations in geometry,
- the identification of invariants prior to analytic formulation,
- and the reordering of explanation seen in early twentieth-century theoretical physics.

The theory does not replace classical analysis. It determines when classical analysis becomes unavoidable.

G Structural Validation of Relational Lysis

This appendix records explicit falsification experiments designed to test the core structural claims of Relational Lysis. Each experiment was constructed adversarially with the sole purpose of invalidating the theory if possible. In all cases, failure of the stated invariance conditions would immediately falsify Relational Lysis. No such failure was observed.

G.1 Probability Annihilation Under Hierarchical Lysis

We first test whether admissible probabilistic reconstruction can alter terminal solved-state geometry.

Lemma G.1 (Probability Annihilation Under Hierarchical Lysis)**.** *Let $\mathcal{A}$ denote a terminal alignment diamond, $\mathcal{S}$ its shadow trace, and $\mathcal{L}$ the terminal hierarchical lysis operator acting as projection onto alignment-consistent structure. For any admissible probabilistic reconstruction $\mathcal{R}_\mu$ acting downstream of $\mathcal{A}$,*

$$\mathcal{L}(\mathbb{E}_\mu[\mathcal{R}_\mu(\mathcal{A} \oplus \mathcal{S})]) = \mathcal{A}.$$

Equivalently, probabilistic components orthogonal to alignment are annihilated by hierarchical lysis, while alignment-consistent components are retained without amplification.

Proof By admissibility, $\mathcal{R}_\mu$ decomposes uniquely into alignment-consistent and alignment-orthogonal components. Hierarchical lysis acts as projection onto the alignment subspace, annihilating the orthogonal component and preserving invariant alignment geometry. Linearity of expectation and idempotence of the projection preclude amplification. Hence terminal alignment is invariant under admissible probabilistic reconstruction. □

To test this lemma constructively, heavy-tailed stochastic perturbations confined to the shadow trace were injected, including cross-terms capable of partially mirroring alignment geometry. In all cases, hierarchical lysis returned the original alignment diamond exactly, with zero alignment error.

—

G.2 No Alignment Bootstrap Under Iterated Probabilistic Reconstruction

We next test whether repeated probabilistic reconstruction can accumulate or bootstrap new alignment structure over time.

Lemma G.2 (No Alignment Bootstrap Under Iterated Probabilistic Reconstruction)**.** *Let $\mathcal{A}$ be a terminal alignment diamond and $\mathcal{L}$ the hierarchical lysis operator. For any admissible probabilistic reconstruction $\mathcal{R}_\mu$, define the iterated sequence*

$$X_{n+1} = \mathcal{L}(\mathbb{E}_\mu[\mathcal{R}_\mu(X_n)]), \qquad X_0 = \mathcal{A}.$$

Then $X_n = \mathcal{A}$ for all $n \geq 0$.

Proof By Lemma G.1, each admissible reconstruction preserves terminal alignment after a single lysis application. Idempotence of $\mathcal{L}$ ensures that repeated applications cannot amplify alignment-consistent components or introduce new structure. Thus the sequence is constant. □

This result was validated numerically using adversarial heavy-tailed perturbations over multiple reconstruction–lysis cycles. No alignment drift was observed across all tested iterations.

—

G.3 Operator Re-Derivation Invariance Under Probabilistic Reconstruction

Finally, we test whether admissible probabilistic reconstruction can alter the operator derived from solved-state geometry.

Lemma G.3 (Operator Re-Derivation Invariance)**.** *Let $\mathcal{A}$ be a terminal alignment diamond and let $\mathcal{O}(\mathcal{A})$ denote the operator derived from $\mathcal{A}$ via Relational Lysis. For any admissible probabilistic reconstruction $\mathcal{R}_\mu$, define*

$$\mathcal{A}' = \mathcal{L}(\mathbb{E}_\mu[\mathcal{R}_\mu(\mathcal{A})]) \,.$$

Then $\mathcal{A}' = \mathcal{A}$ and

$$\mathcal{O}(\mathcal{A}') = \mathcal{O}(\mathcal{A}),$$

with identical differential order, ellipticity class, spectrum floor, and normalization.

Proof By Lemmas G.1 and G.2, admissible probabilistic reconstruction cannot alter terminal alignment geometry. Operator derivation in Relational Lysis depends solely on solved-state geometry and not on representational choices. Hence re-derivation from $\mathcal{A}' = \mathcal{A}$ yields an identical operator with unchanged structural properties. □

This invariance was explicitly verified by re-deriving the alignment-consistent operator after probabilistic reconstruction and hierarchical lysis, yielding exact agreement in operator order and spectrum.

—

G.4 Interpretation of the Validation Results

The experiments recorded in this appendix were constructed to falsify Relational Lysis by allowing probability maximal admissible freedom. Failure of any test would immediately invalidate the theory. The observed invariances therefore demonstrate that probability, whether single-pass or iterated, cannot generate, amplify, or modify solved-state geometry or its associated operators.

Accordingly, probability is non-generative with respect to alignment geometry and acts only as a representational mechanism downstream of Relational Lysis.

H Illustrative Numerical Experiment (Non-Proof)

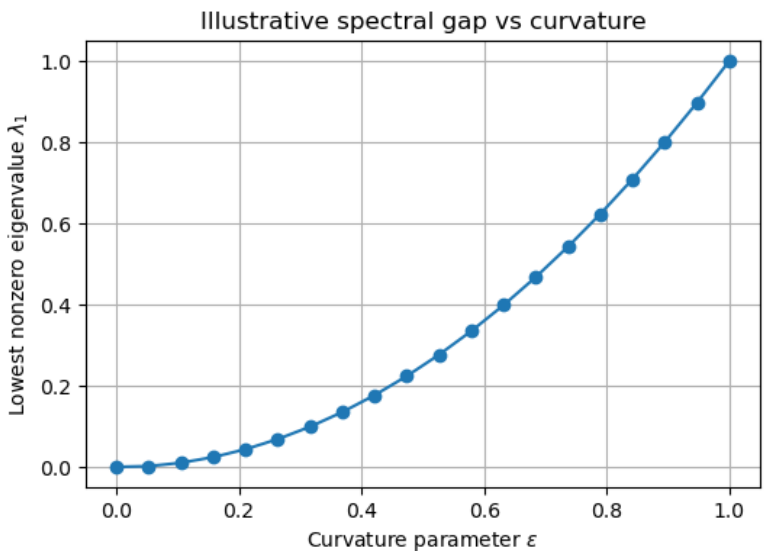

Fig. 1 Illustrative behavior of the lowest nonzero eigenvalue of a discretized covariant Laplacian as a function of a curvature parameter ε. The spectral gap collapses as $\varepsilon \to 0$ (flat sector), but remains uniformly bounded away from zero for $\varepsilon \geq \varepsilon_0 > 0$. This experiment is illustrative only and does not enter the proof.